

Topological Data Analysis

Lecture 16

Hierarchical clustering

Oleg Kachan

Clustering



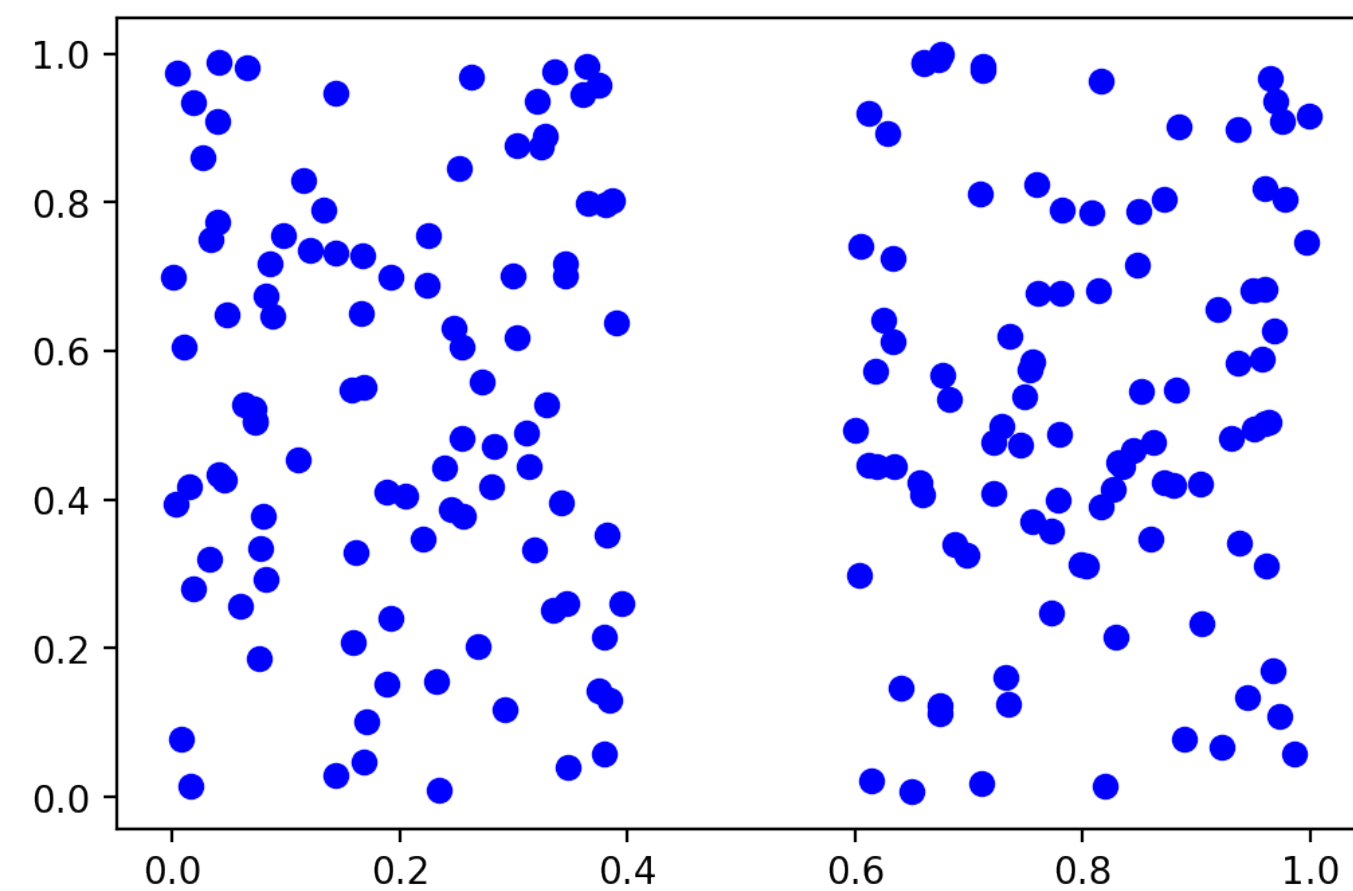
Given a set X , a clustering can be posed as a partition of X , that is

$$X = \{C_1, \dots, C_n\}$$

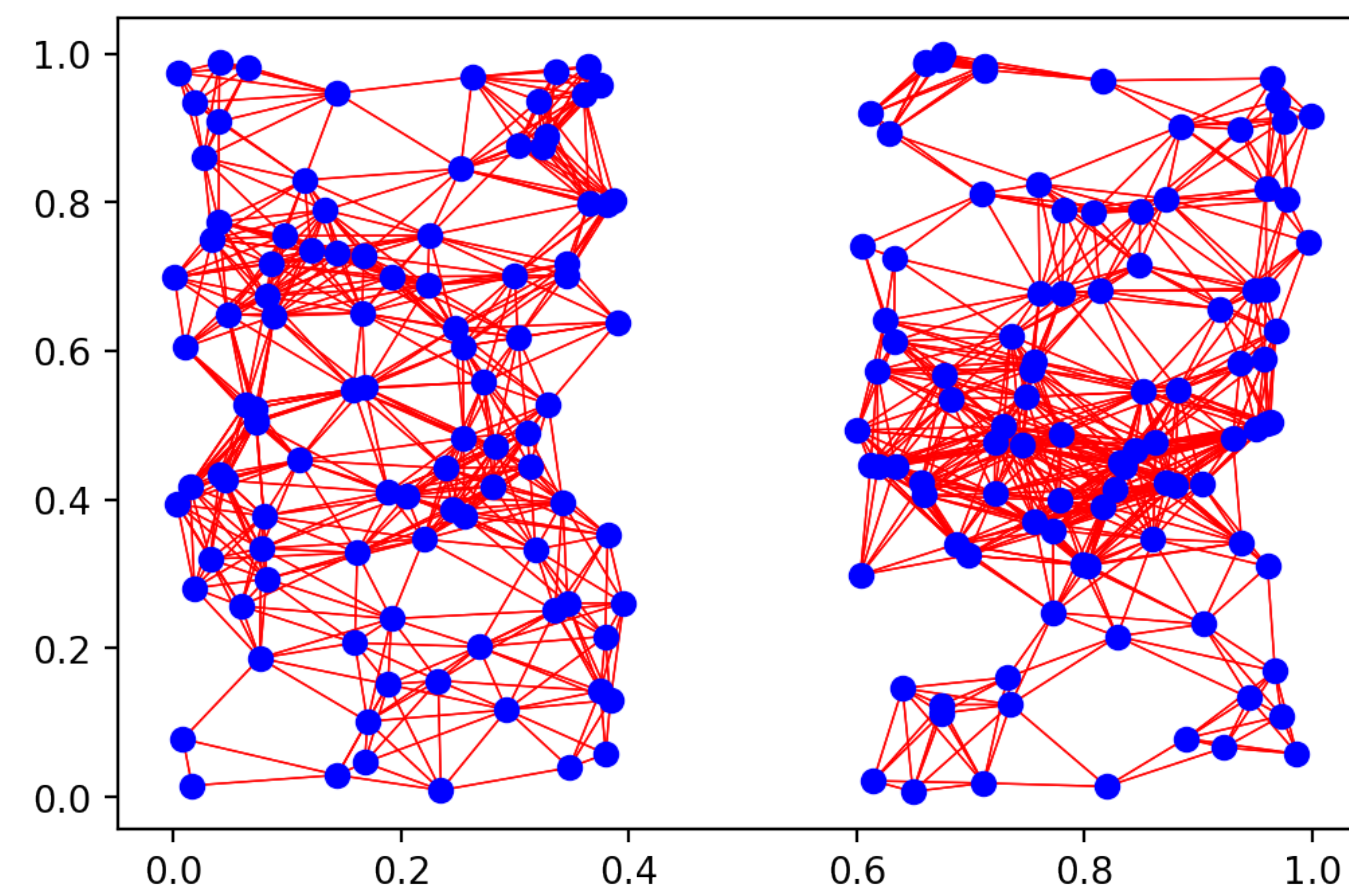
such that

$$X = \bigcup_{i=1}^n C_i \quad C_i \cap C_j = \emptyset, \forall i, j$$

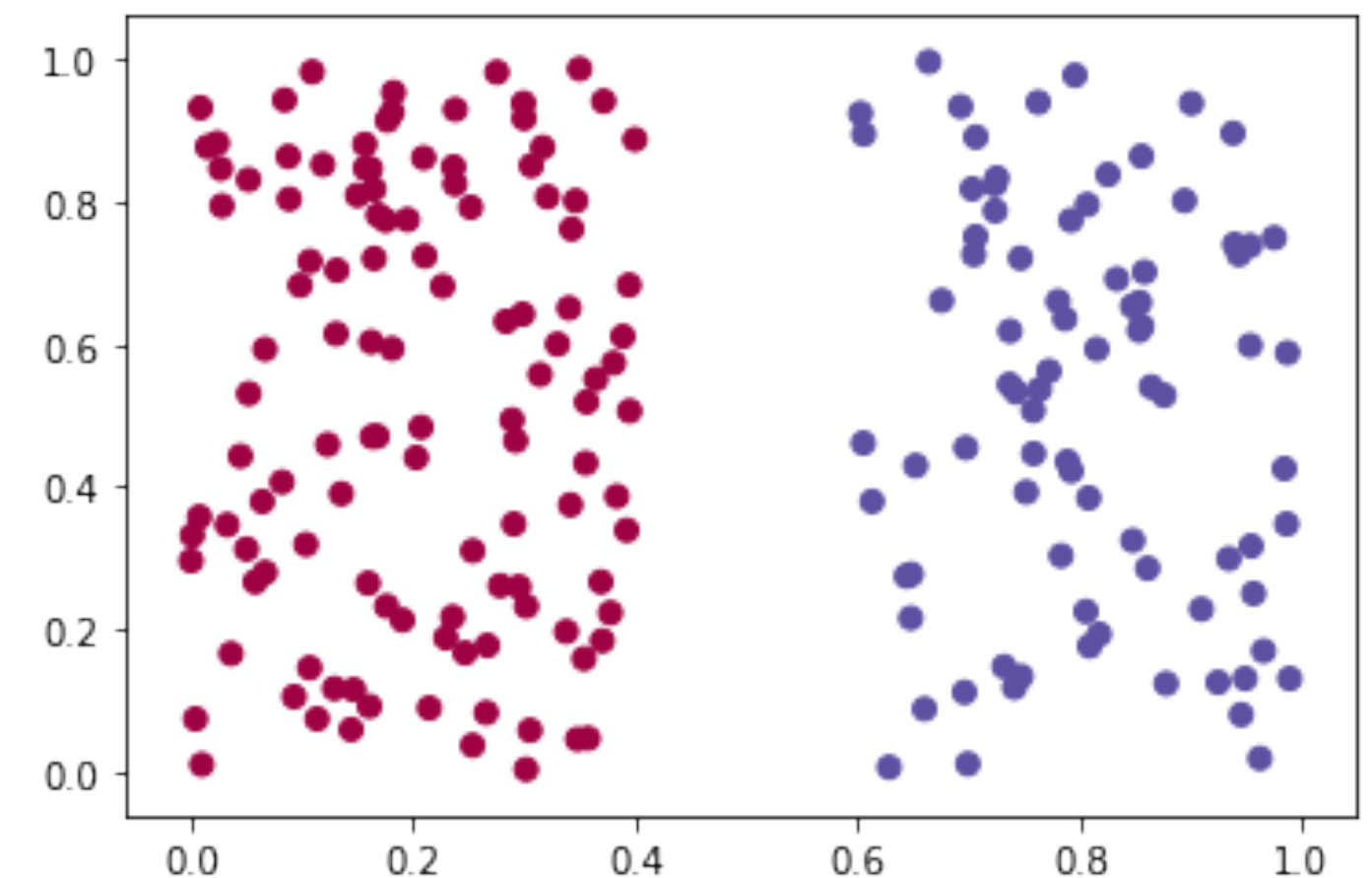
Clustering as connected components



X



$G_t(X)$

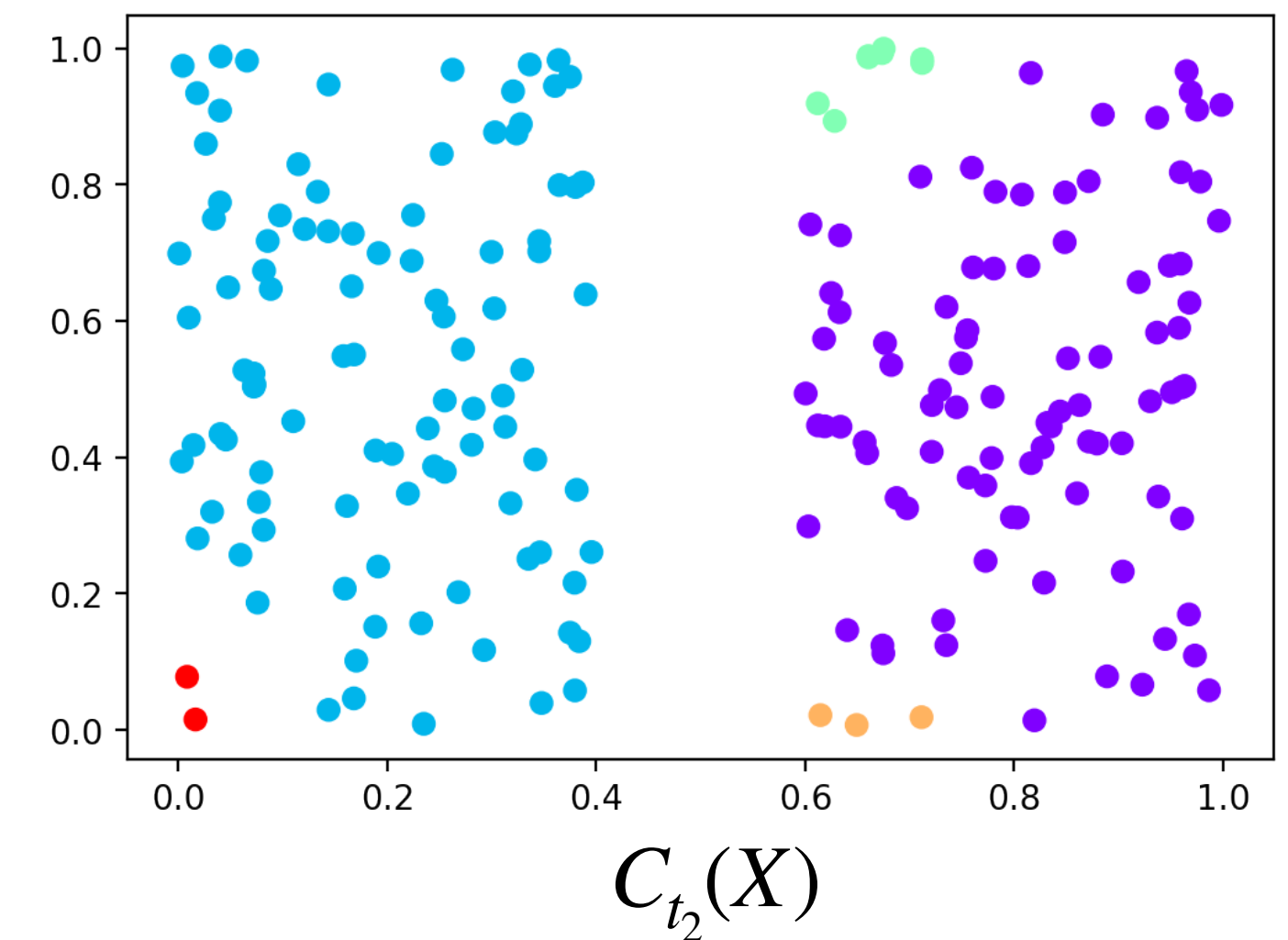
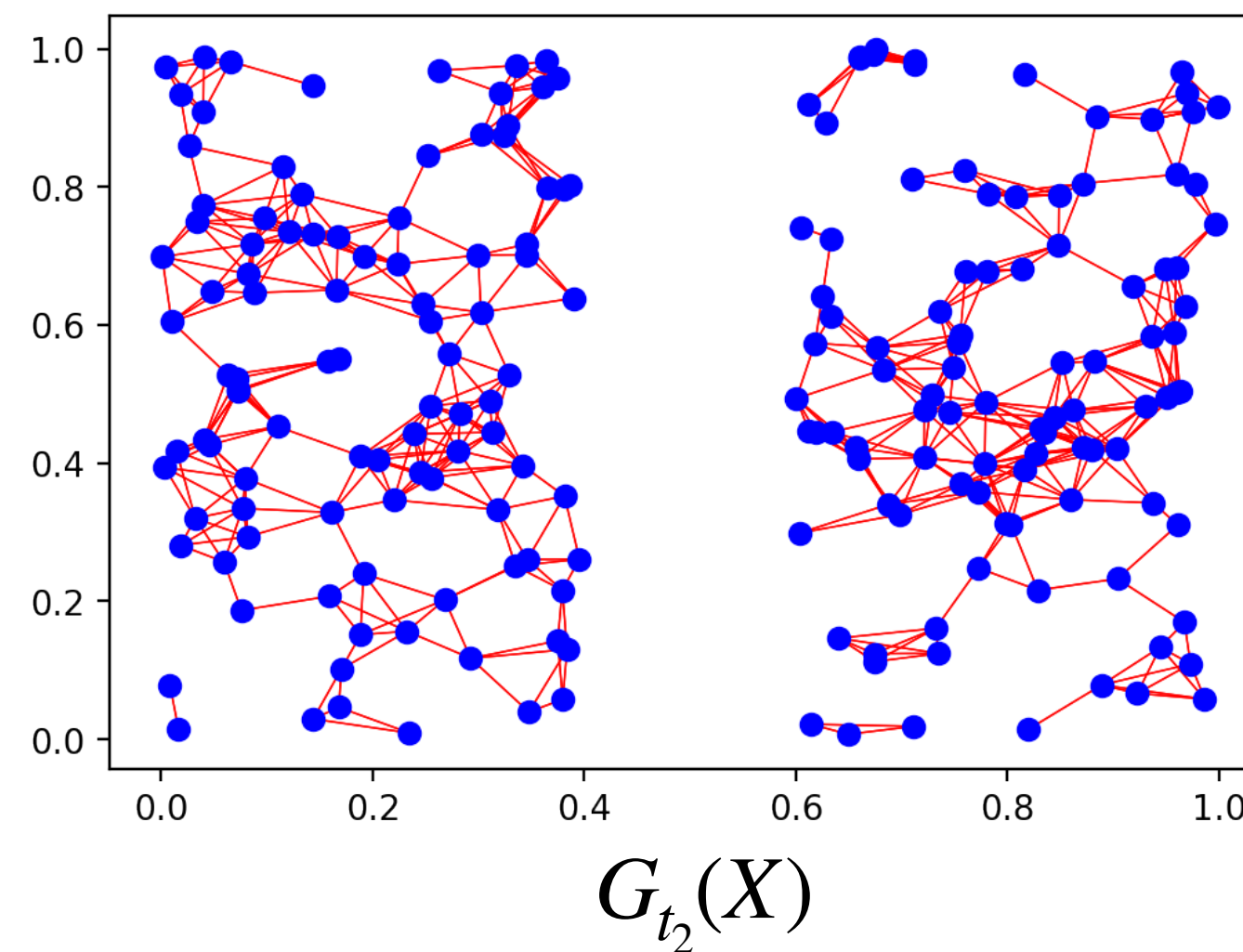
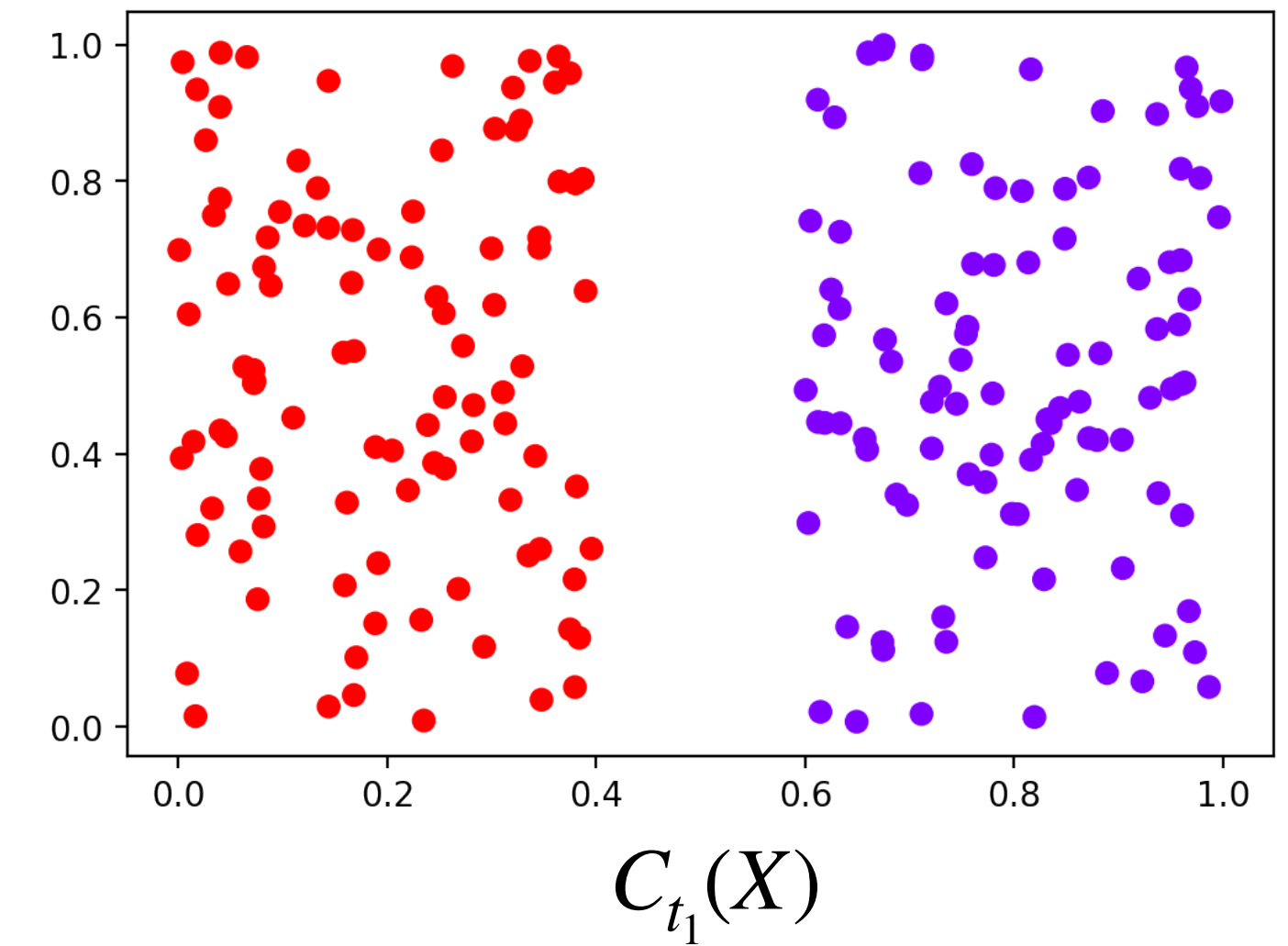
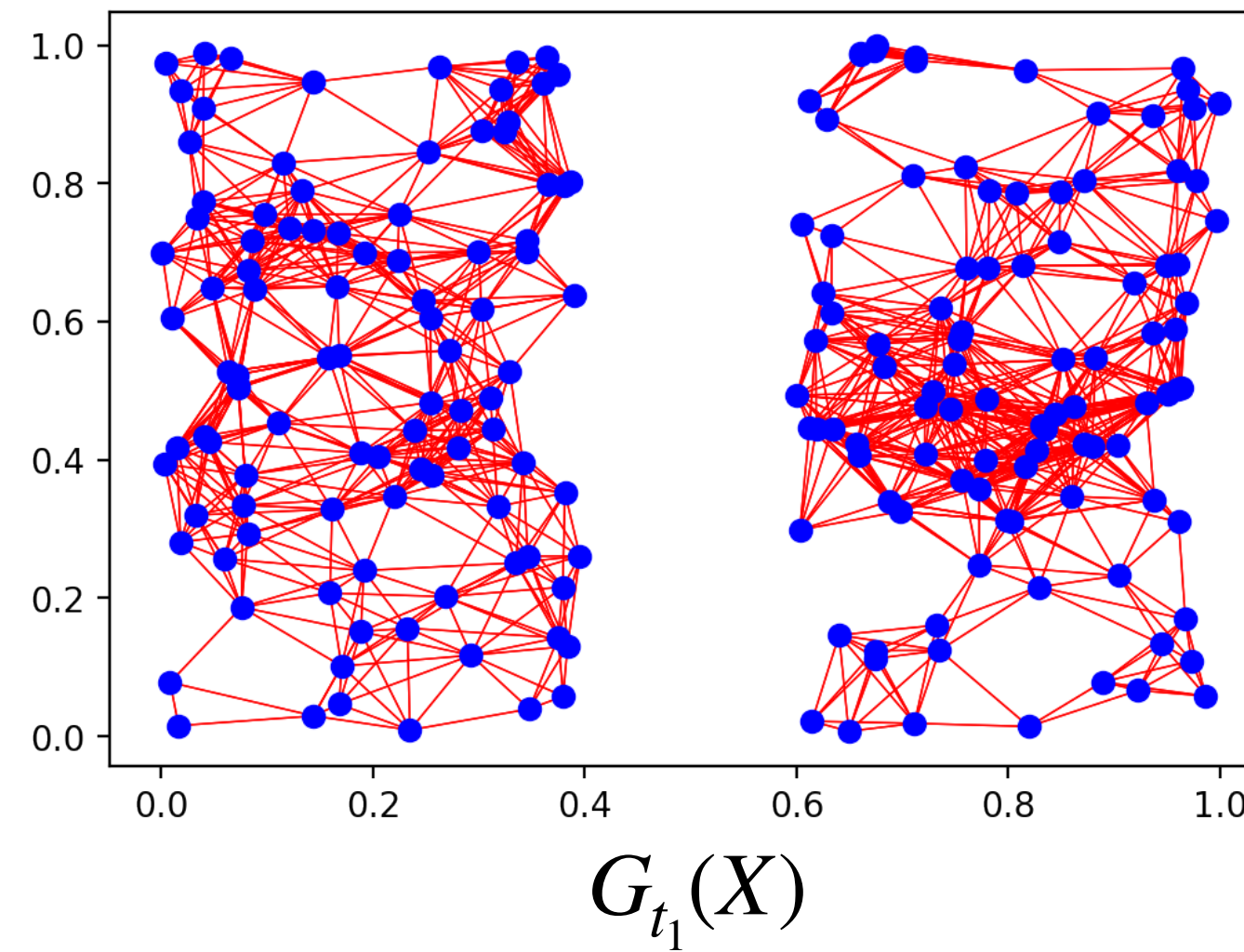
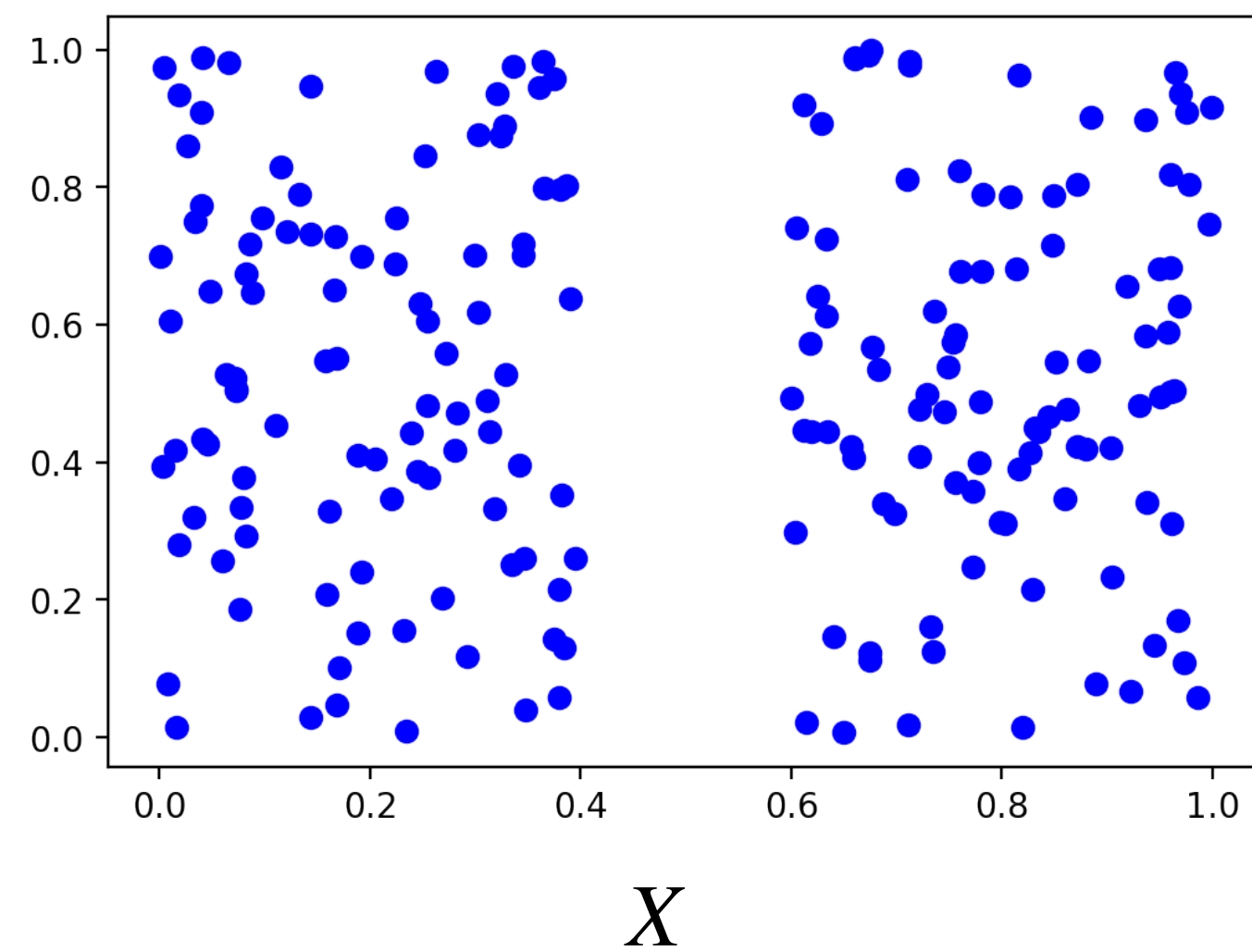


$C_t(X)$

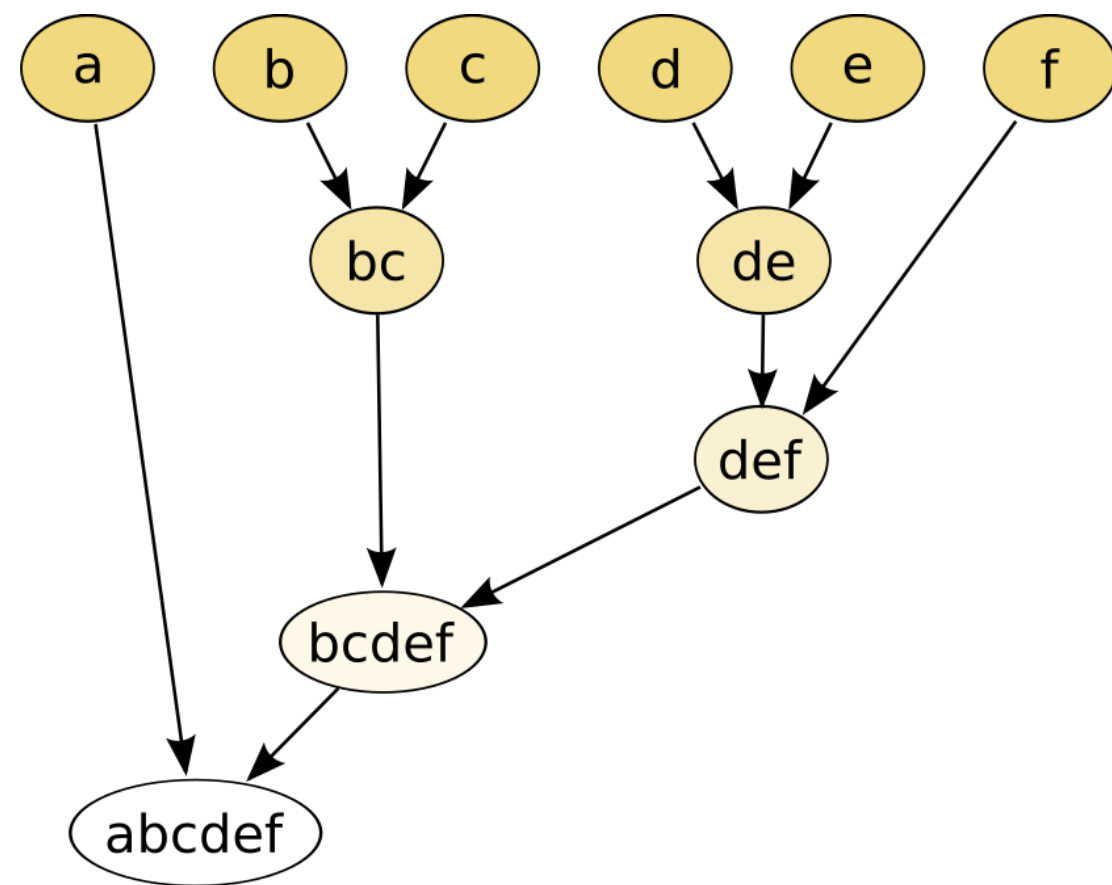
- clusters — 0-th homology classes of the neighborhood graph of X
- kNN-, eps-ball graphs, etc.

Clustering as connected components

How to select the threshold parameter?



Hierarchical clustering



$$\mathcal{C} = \{C_1, \dots, C_n\}$$

Hierarchical structure

If $C, C' \in \mathcal{C}$ s.t. $C \neq C'$ then $C \cap C' = \emptyset$, or $C \subset C'$, or $C' \subset C$

Single linkage

$$D(X, Y) = \min_{x \in X, y \in Y} d(x, y)$$

Complete linkage

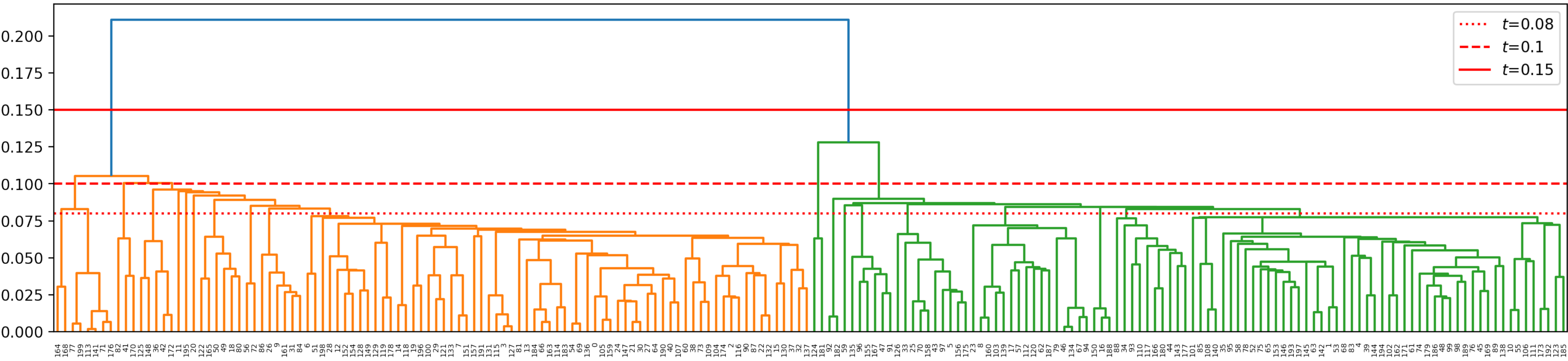
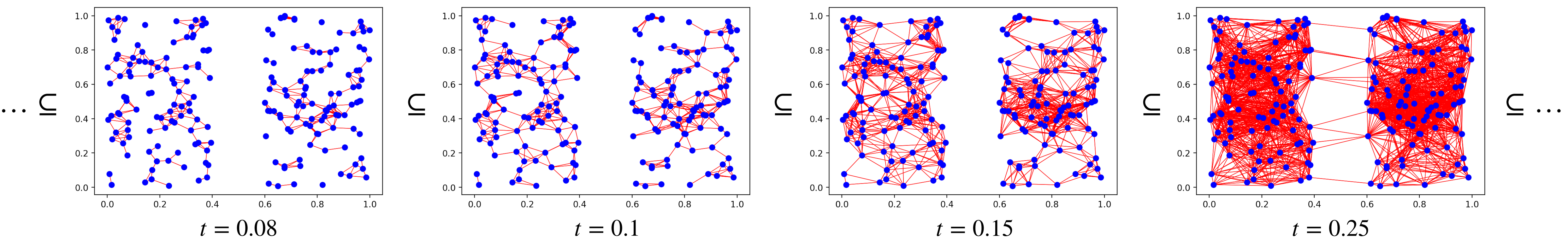
$$D(X, Y) = \max_{x \in X, y \in Y} d(x, y)$$

Average linkage

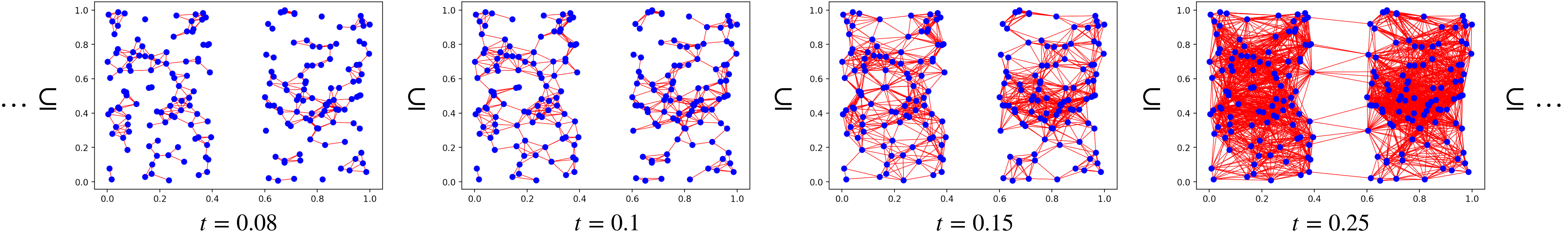
$$D(X, Y) = \frac{1}{|X||Y|} \sum_{x \in X} \sum_{y \in Y} d(x, y)$$

Hierarchical clustering

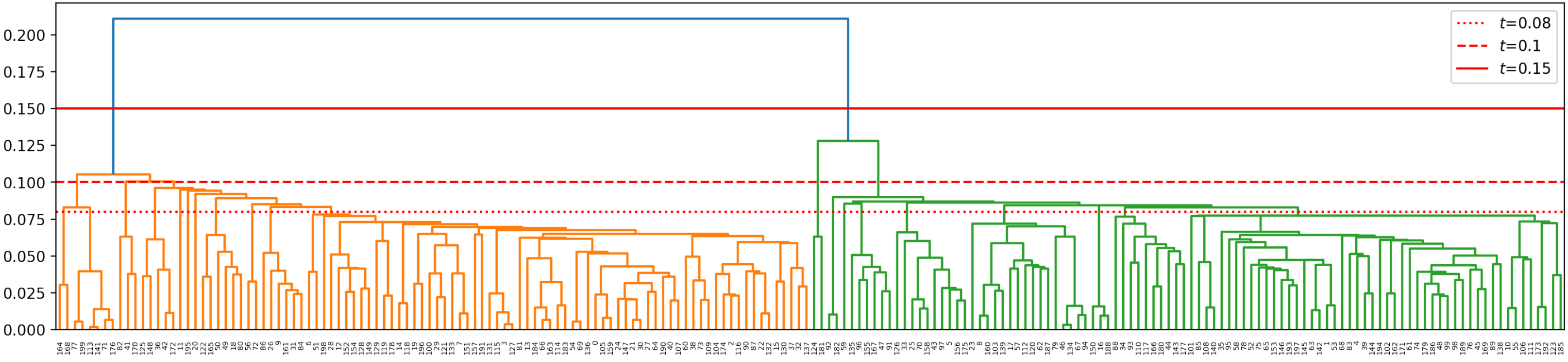
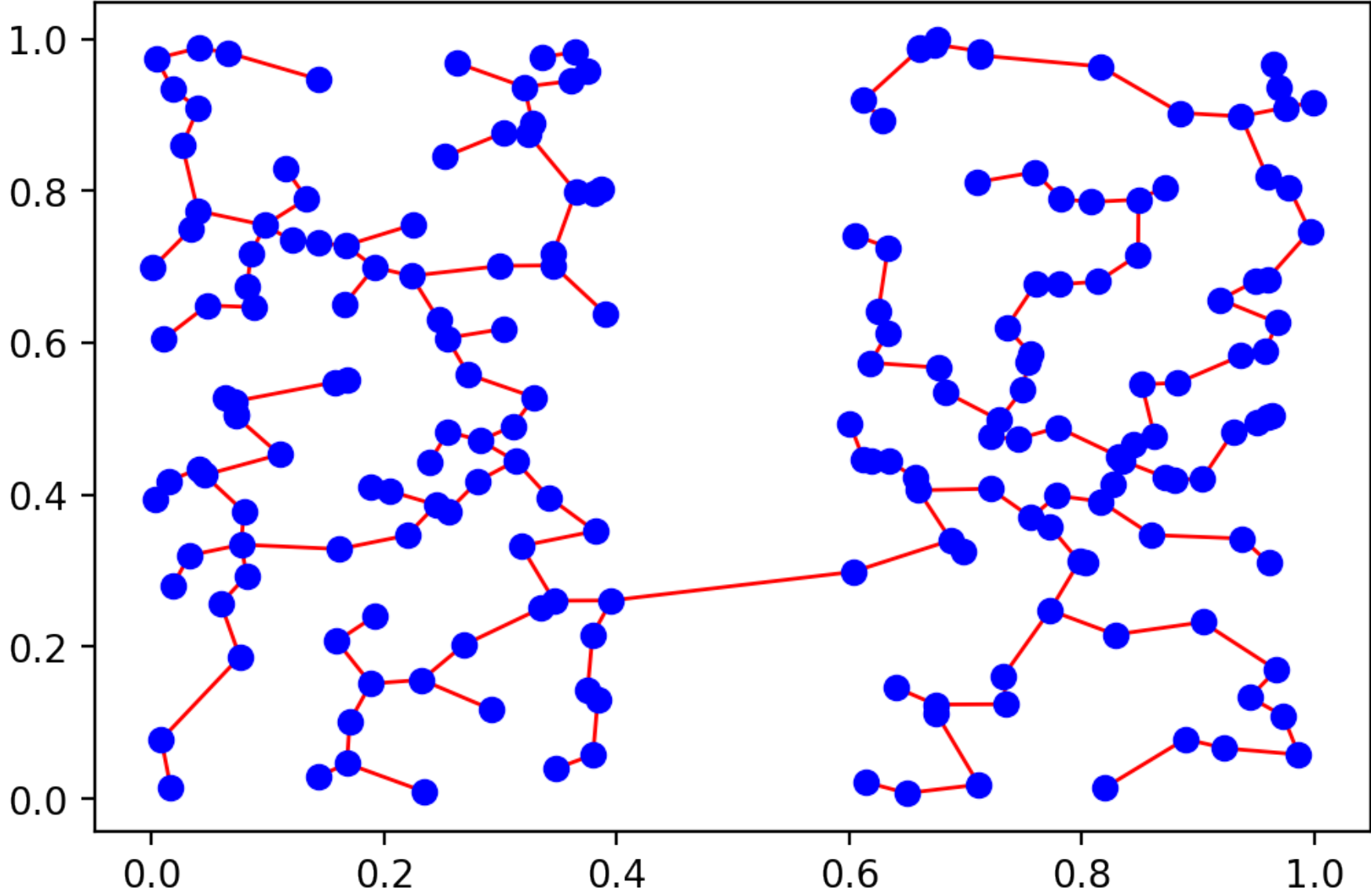
- Perform a filtration over all parameters, clusters at t 0-th homology classes of the neighborhood graph of X



Minimum spanning tree



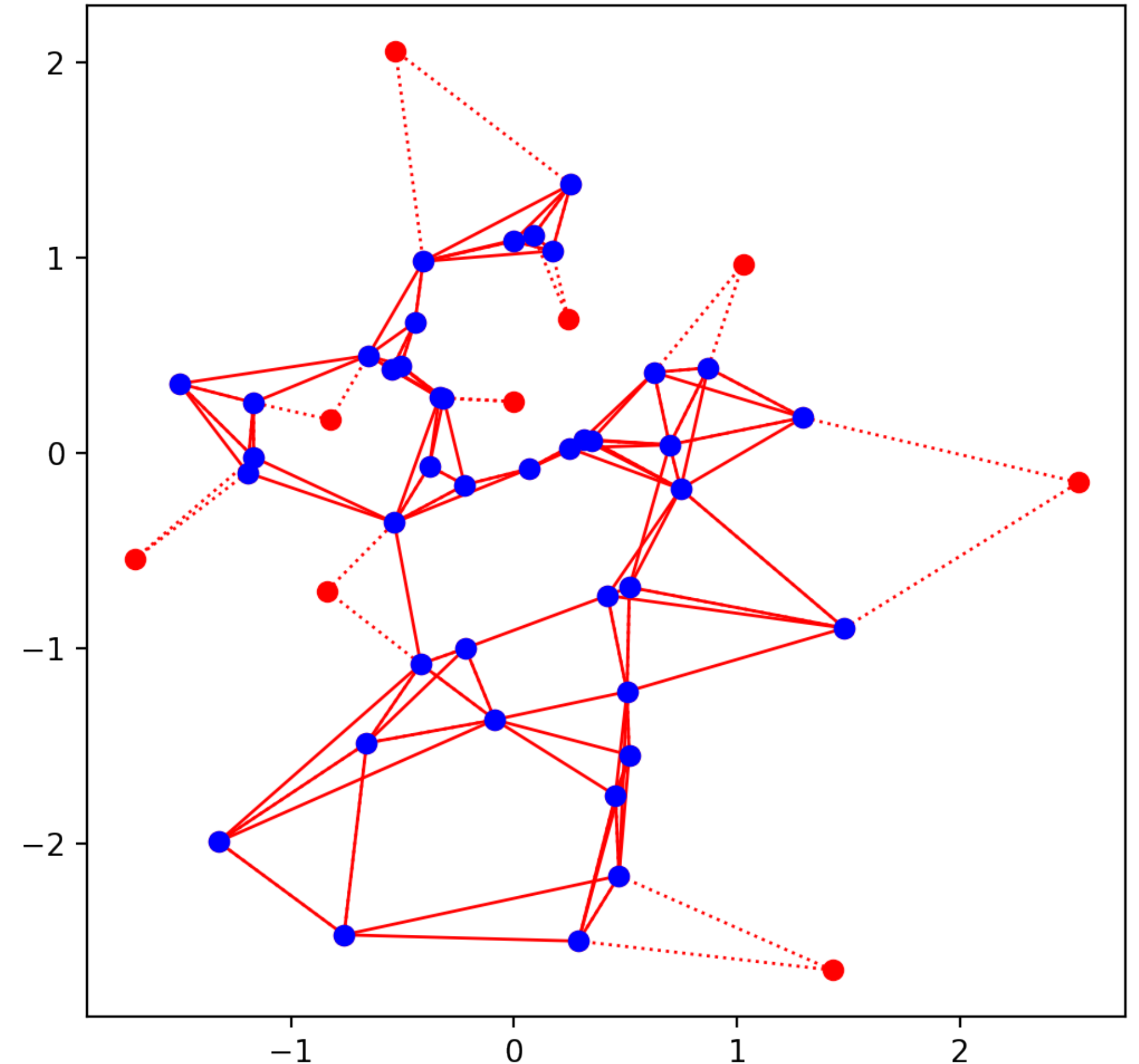
Minimum spanning tree (MST)



Density-based clustering

DBSCAN

- for each data point calculate a density estimate
 - choose a density threshold λ and find the observations with $p(x_i) > \lambda$
 - construct a graph connecting each high density observation to all other observations within distance r
 - define the clusters to be the connected components of this graph
-
- not every point is in cluster (or a separate cluster of outliers)
 - clusters are 0-th homology classes of density-Rips graph, for a fixed density threshold λ

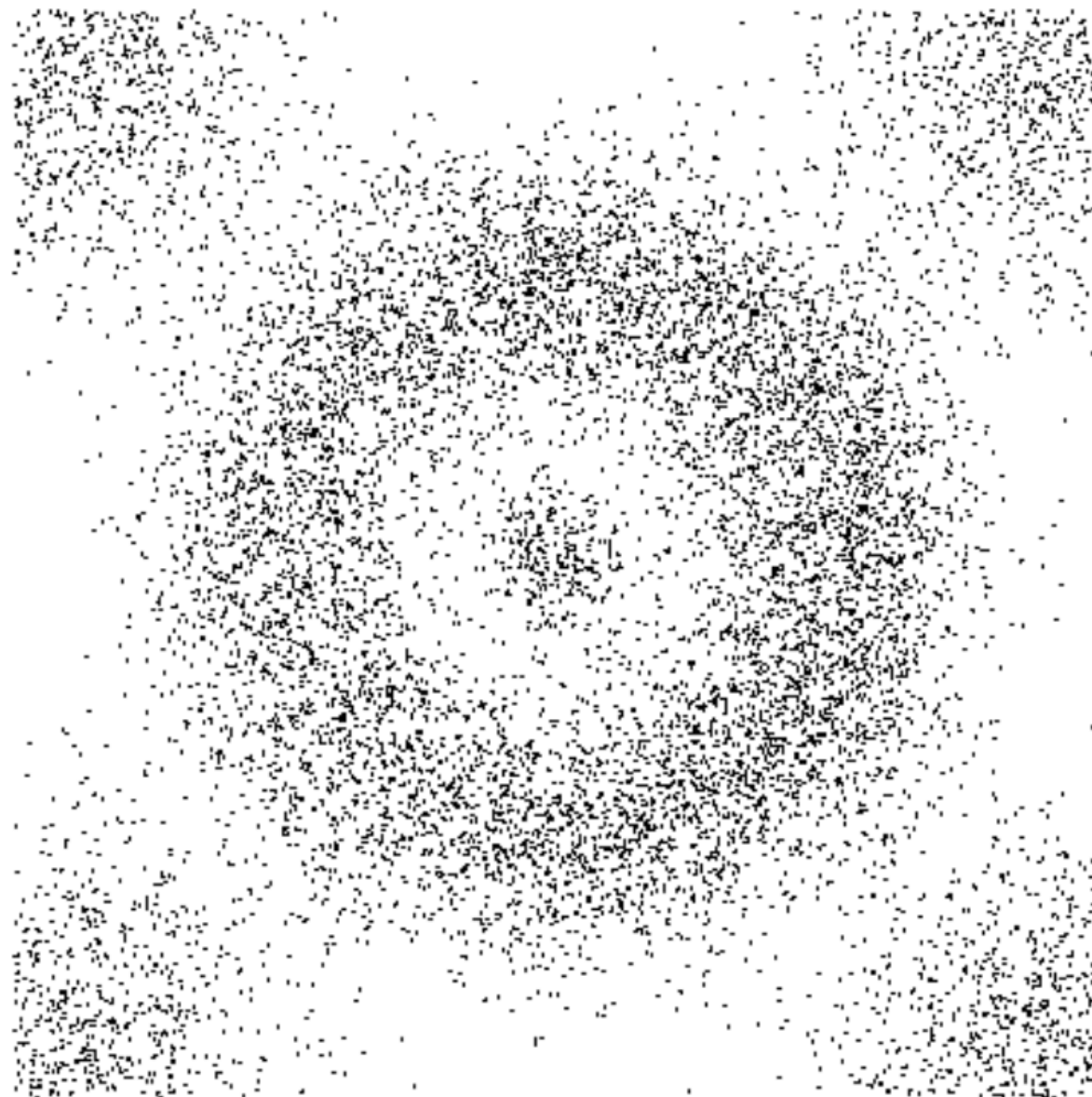


Hierarchical density-based clustering

- **HDBSCAN** — fix density estimate, perform a filtration over distance, prune the cluster tree
- **ToMaTo** — compute a density estimate over a neighborhood graph for fixed ε , perform filtration over density, prune the cluster tree by persistence

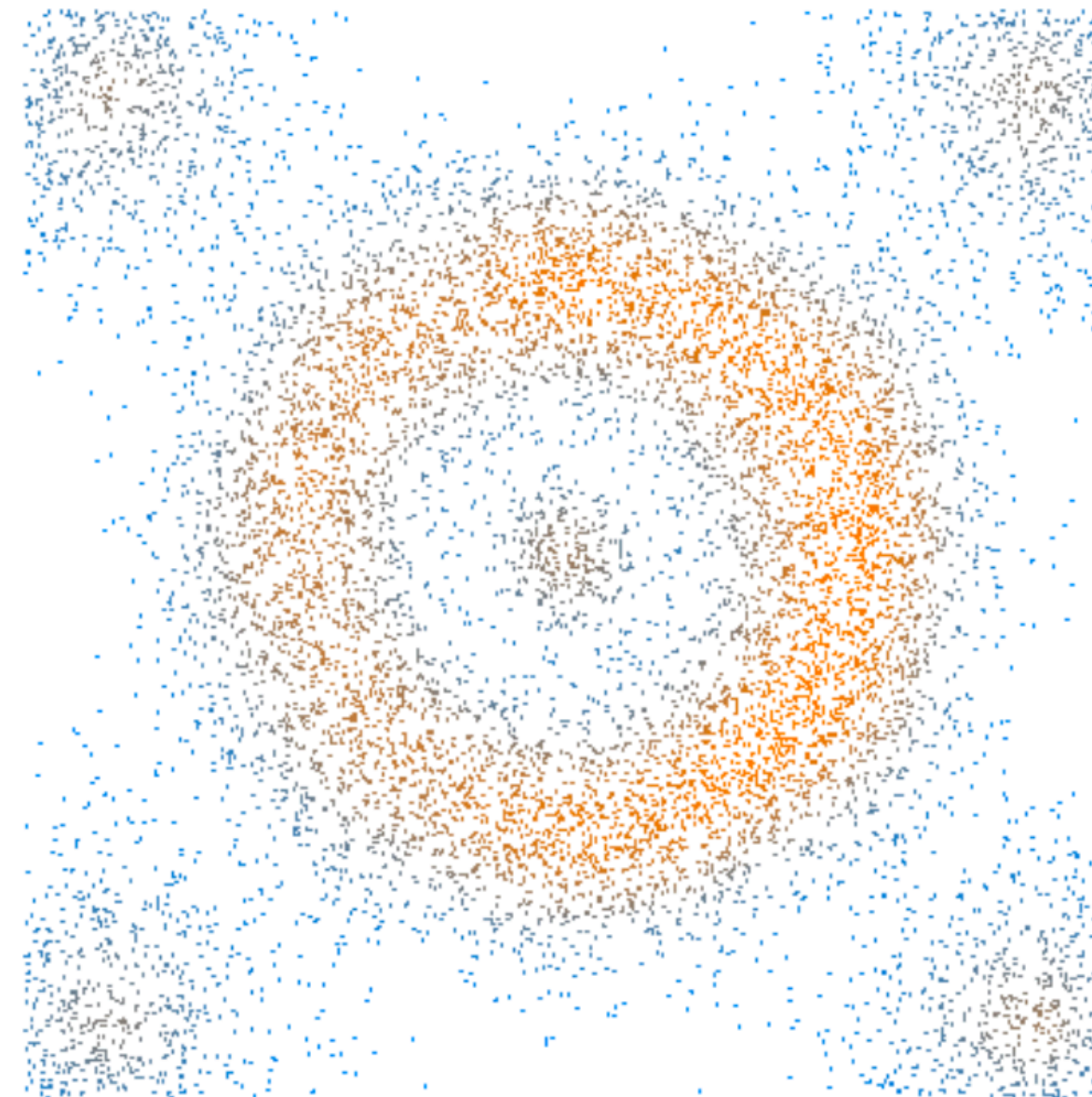
ToMaTo

Input data



X

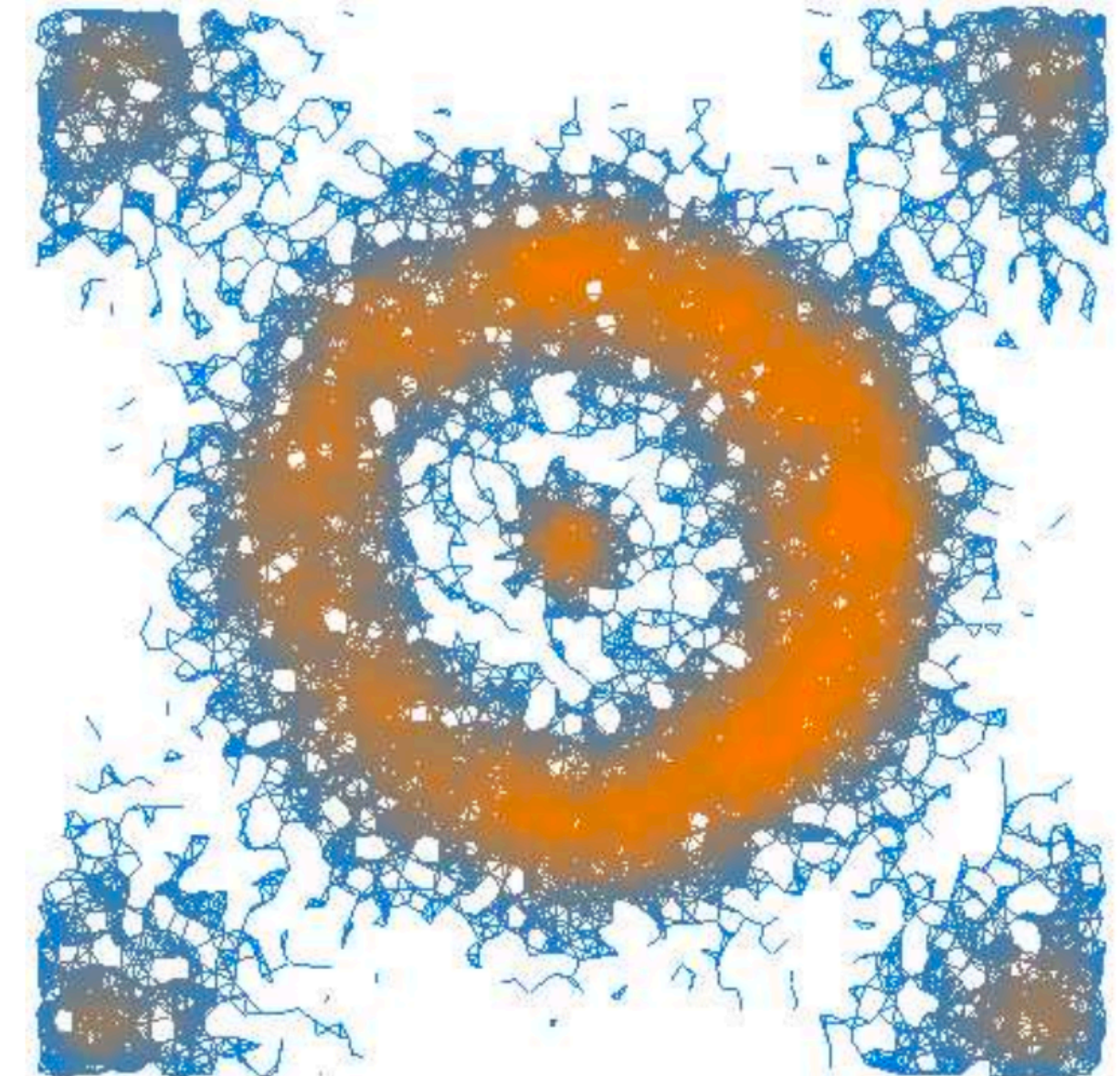
Point density estimate



$p(X)$

Neighborhood graph

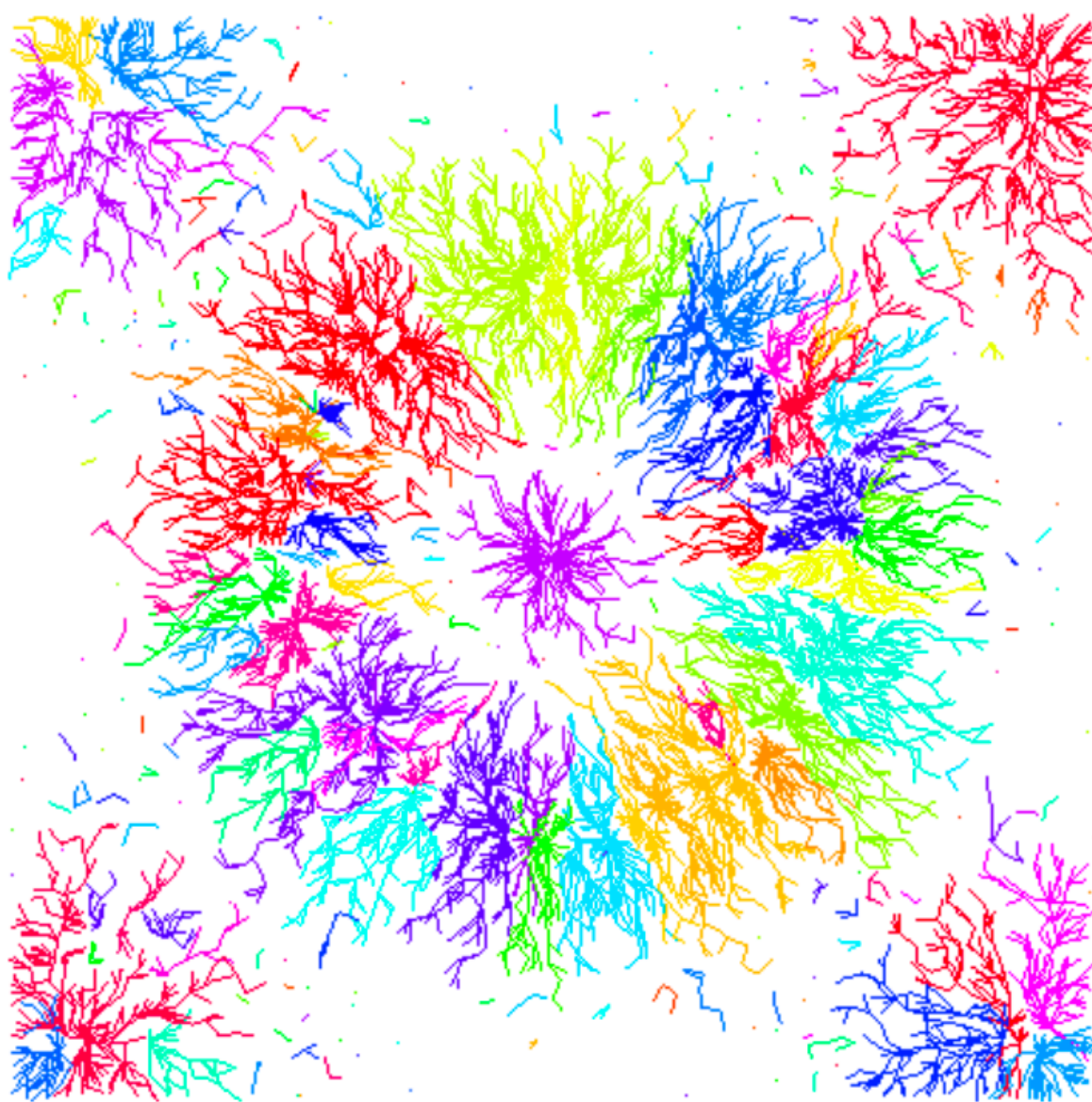
Fixed neighborhood size



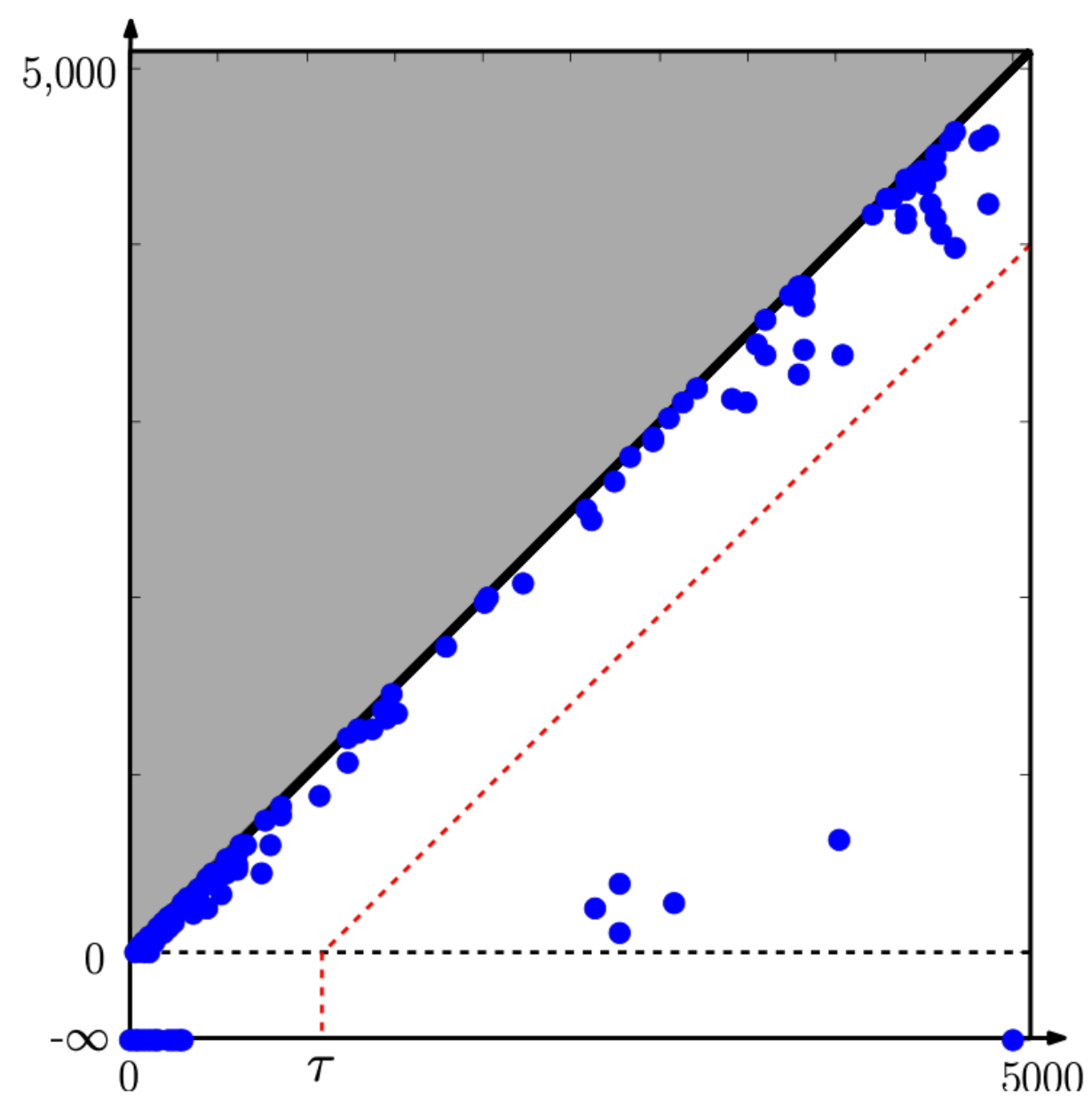
$G_\epsilon(X)$

ToMaTo

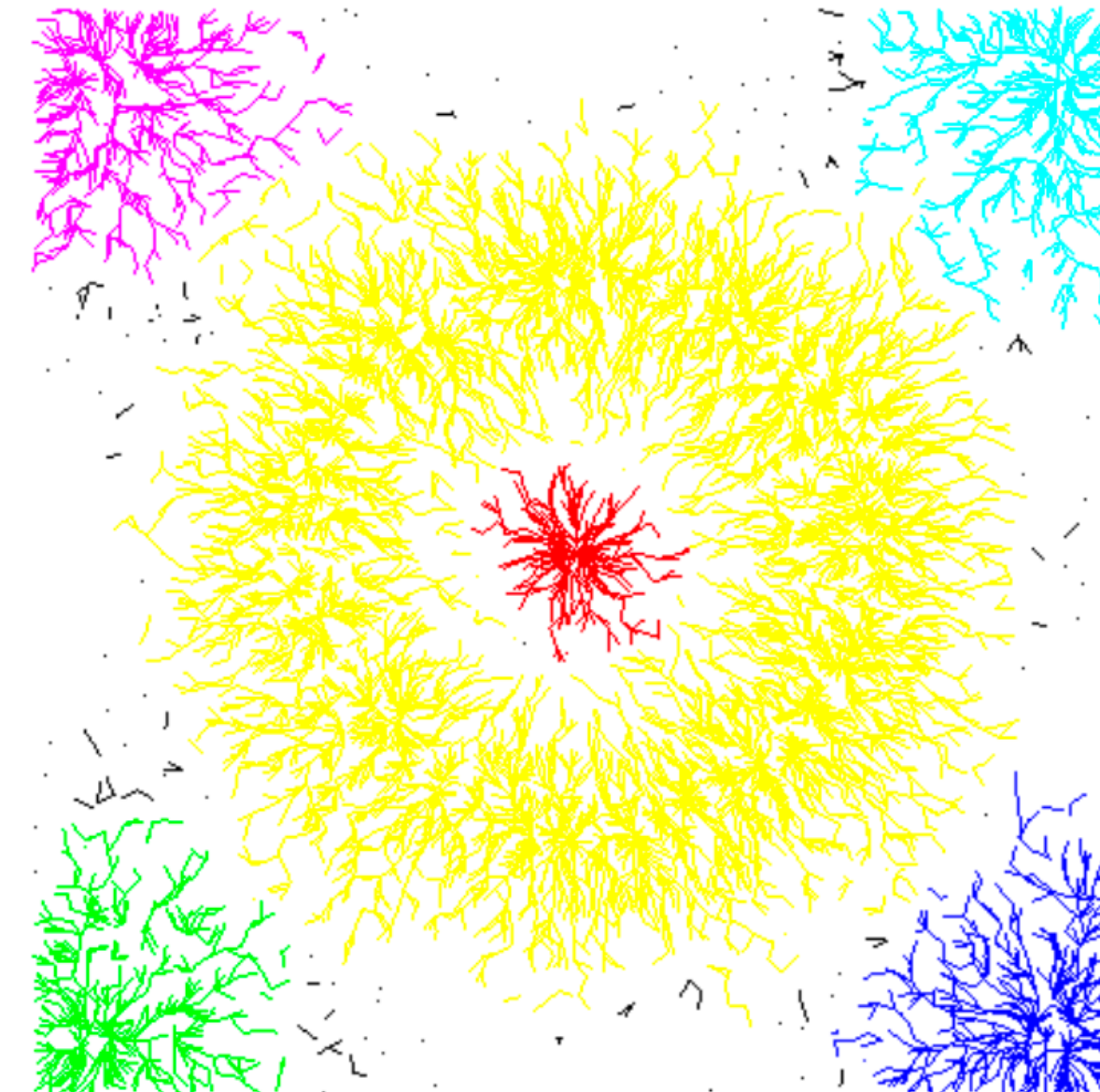
Hill climbing



Persistence

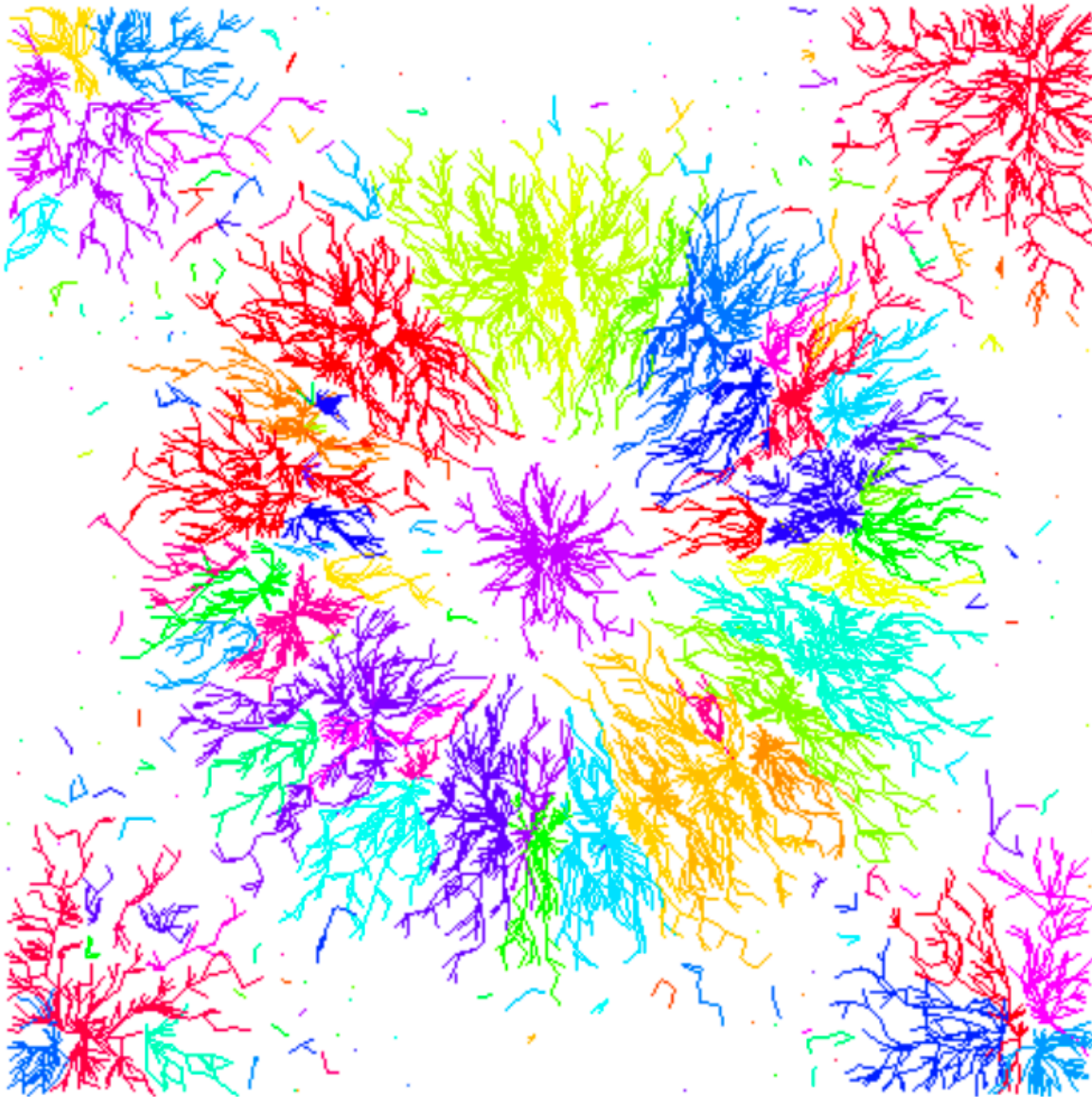


Cluster merging



ToMaTo

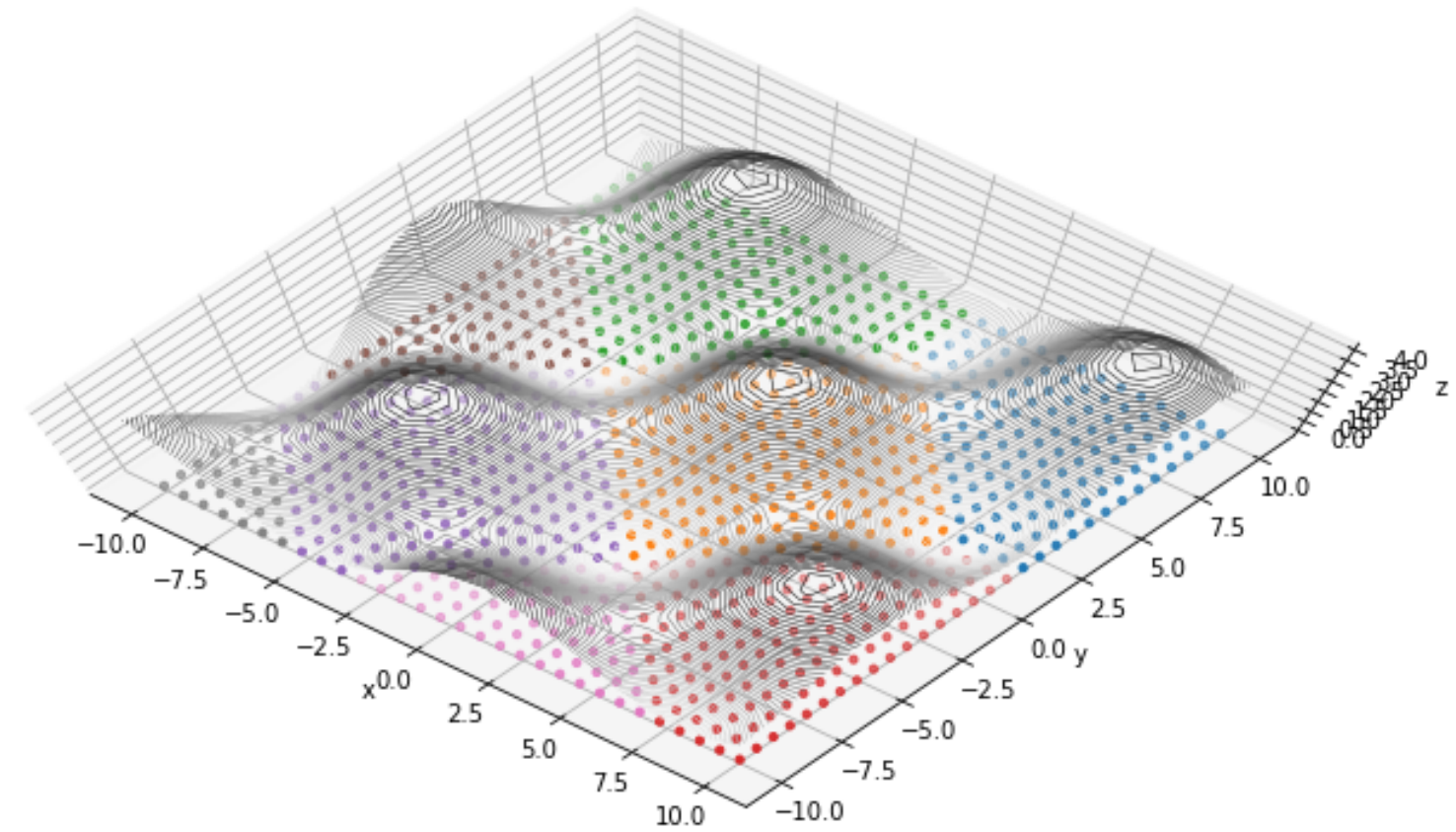
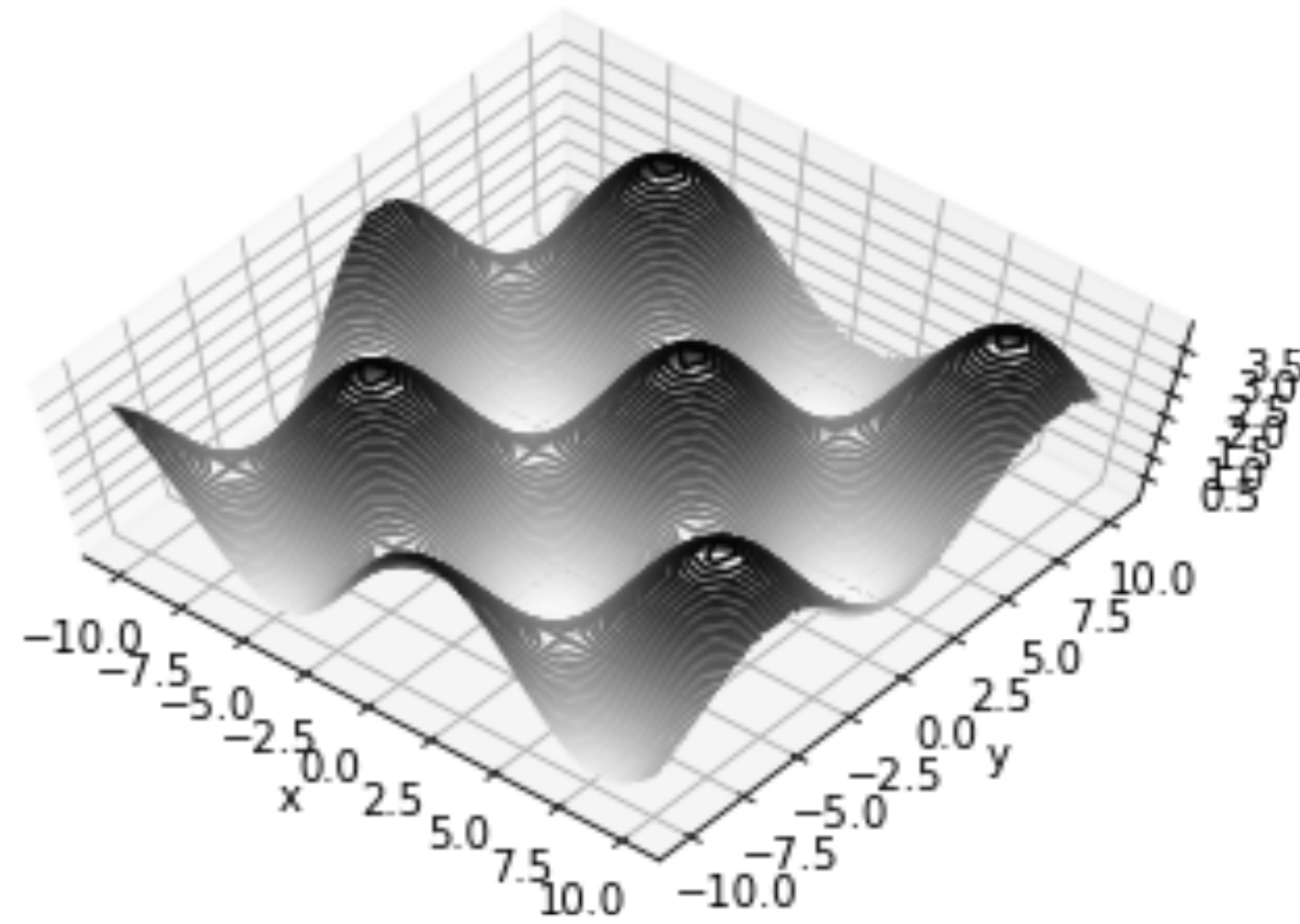
Hill climbing



Apply a hill-climbing algorithm on G . Construct the neighborhood graph G on the points of X , and construct a directed subgraph Γ of G as follows: at each vertex v of G , place a directed edge from v to its neighbor with highest value of $\hat{f}$, provided that that value is higher than $\hat{f}(v)$. If all neighbors of v have lower values, v is a peak of $\hat{f}$. This yields a collection of directed edges that form a spanning forest of the graph G , consisting of one tree for each local maximum of $\hat{f}$. In particular, these trees yield a partition of the elements of X into pairwise disjoint sets that serves as a candidate clustering on X .

^

ToMaTo

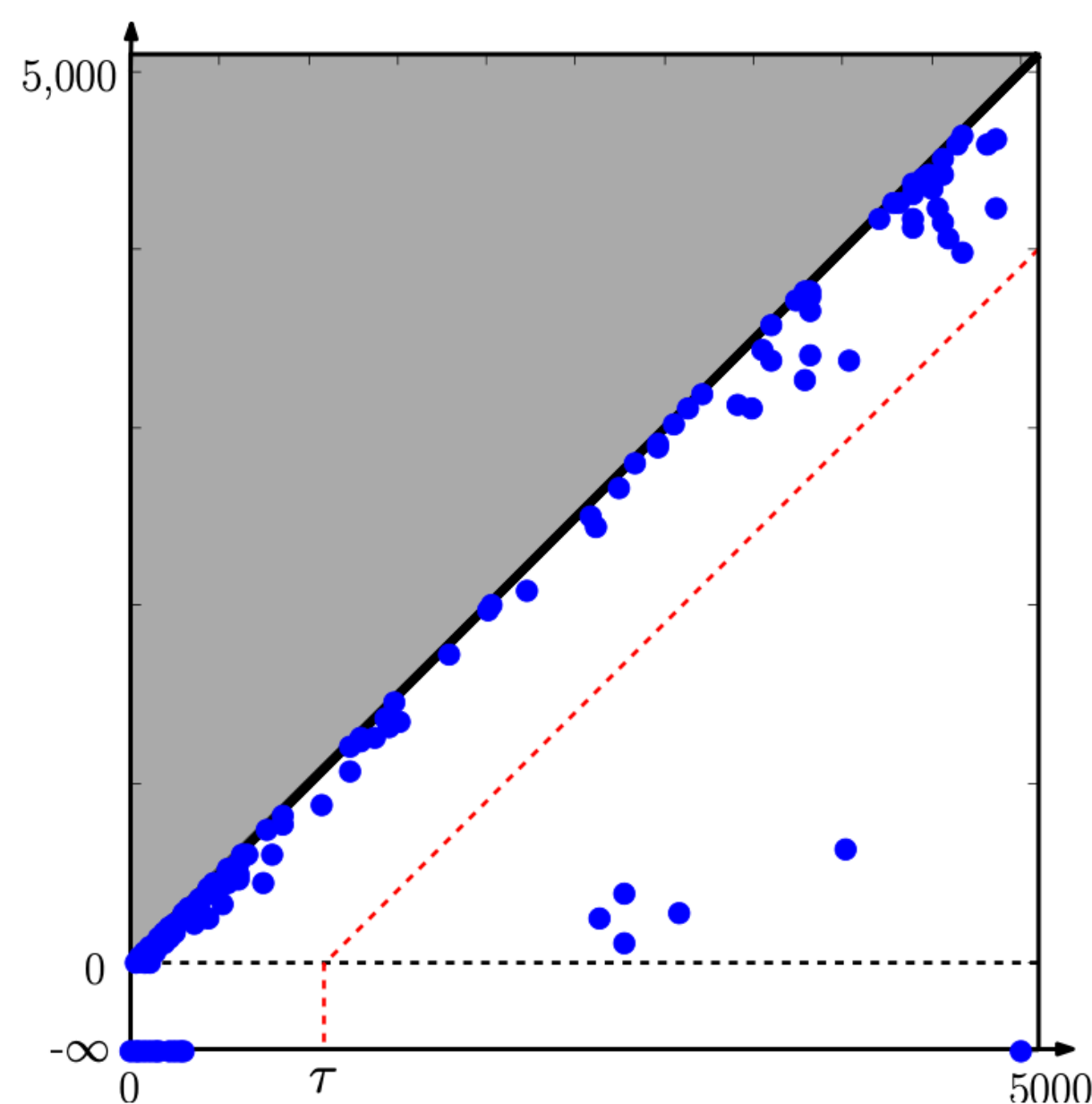


Apply a hill-climbing algorithm on G . Construct the neighborhood graph G on the points of X , and construct a directed subgraph Γ of G as follows: at each vertex v of G , place a directed edge from v to its neighbor with highest value of $\hat{f}$, provided that that value is higher than $\hat{f}(v)$. If all neighbors of v have lower values, v is a peak of $\hat{f}$. This yields a collection of directed edges that form a spanning forest of the graph G , consisting of one tree for each local maximum of $\hat{f}$. In particular, these trees yield a partition of the elements of X into pairwise disjoint sets that serves as a candidate clustering on X .

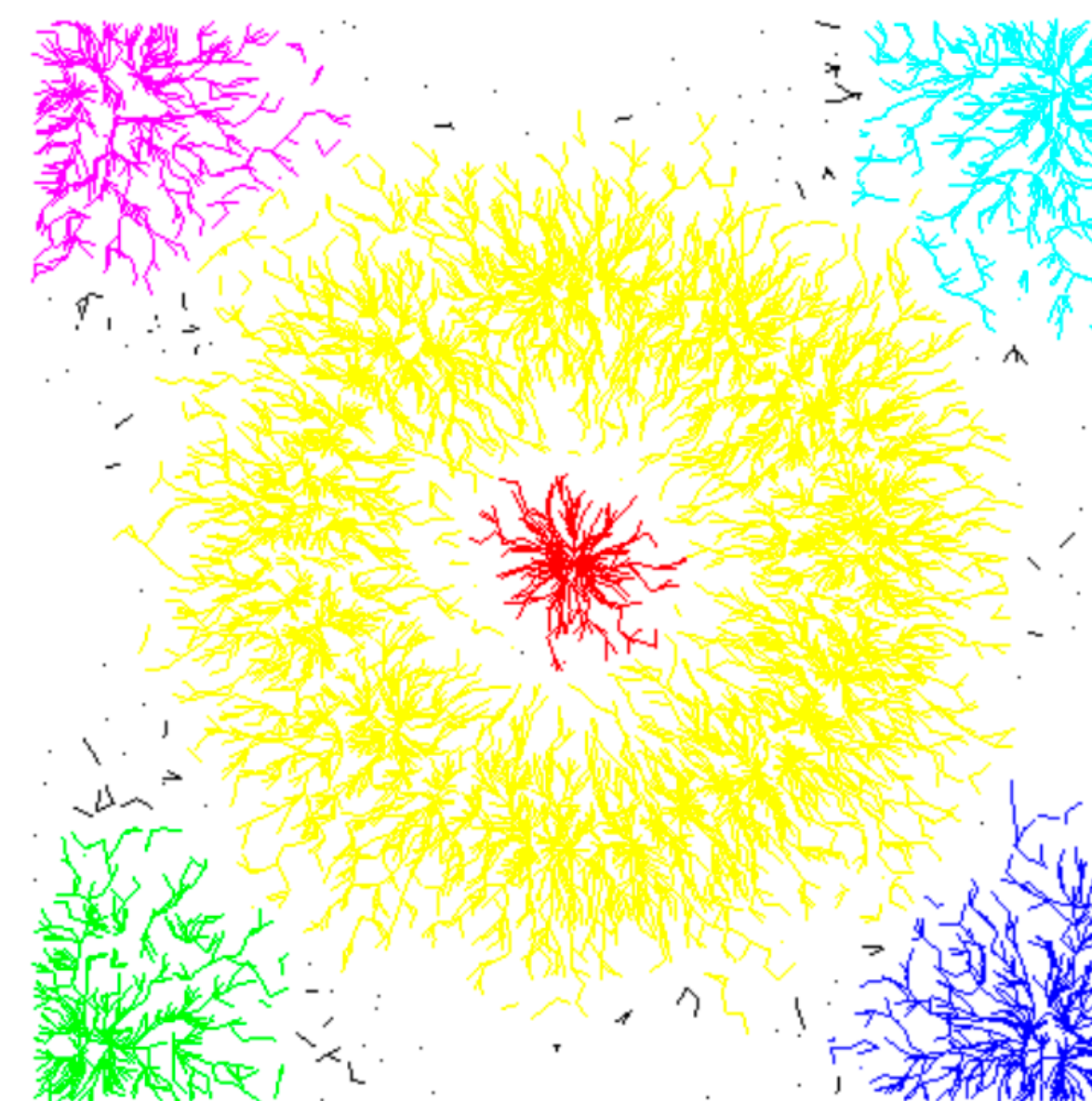
^

ToMaTo

Persistence



Cluster merging



Construct the persistence diagram. Construct the persistence diagram $\text{Dgm}(G, \hat{f})$ associated to the superlevelset filtration of $\hat{f}: G \rightarrow \mathbb{R}$.

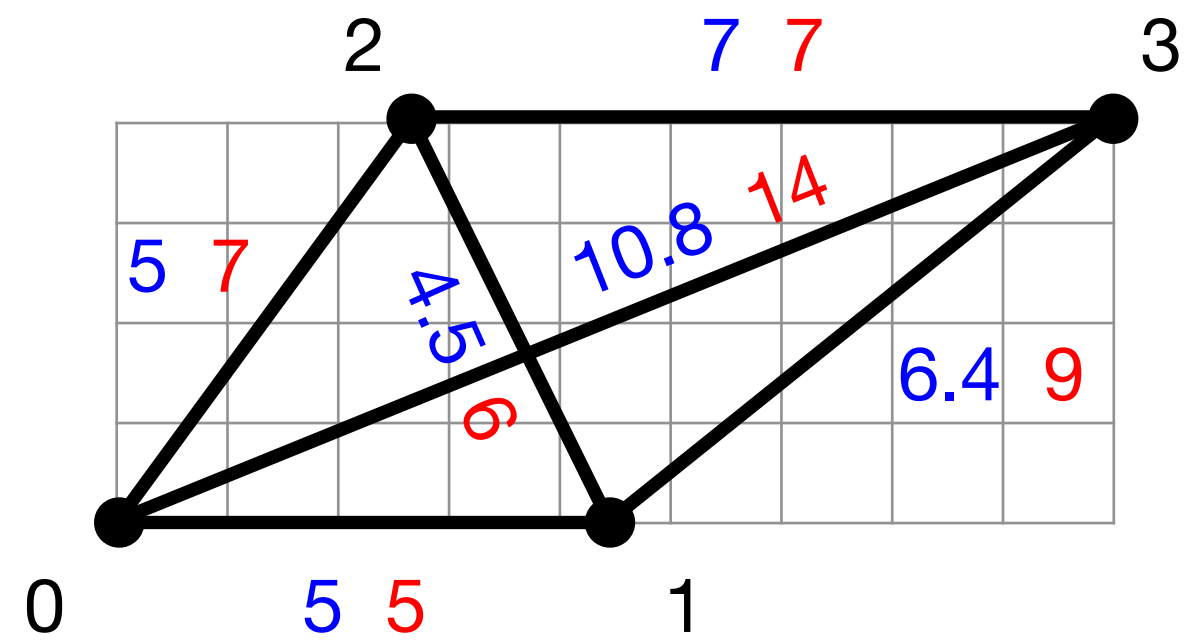
Merge non-significant clusters. Iteratively merge every cluster of prominence less than τ of the candidate clustering found in Step 2 into its parent cluster, i.e., into the cluster corresponding to the local maximum that it gets merged into in the superlevelset filtration of $\hat{f}: G \rightarrow \mathbb{R}$. ToMaTo outputs the resulting clustering of points of X , in which every cluster has prominence at least τ by construction.

Multiparameter hierarchical clustering

- filter by both parameters, get bifiltration
- slice bifiltration to get a filtration, thus a clustering tree
- other function may be different from density, intrinsic dimension

Bifiltration

Euclidean distance filtration

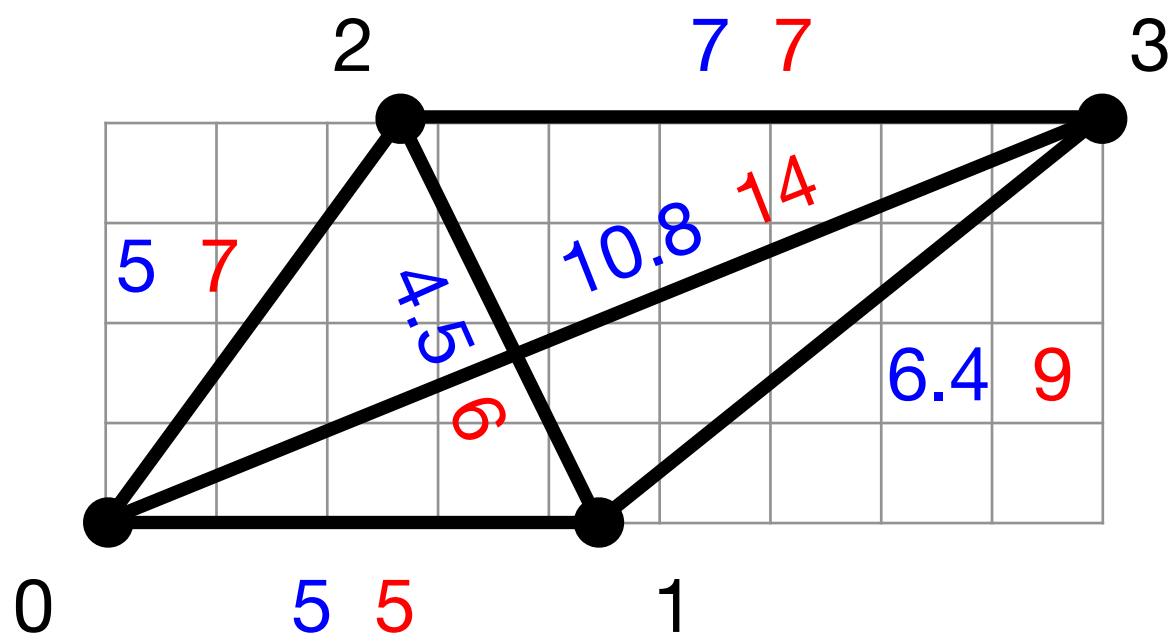


f	σ
0	0
0	1
0	2
0	3
4,5	12
5	01
5	02
6,4	13
7	23
10,8	03

Manhattan distance filtration

f	σ
0	0
0	1
0	2
0	3
5	01
6	12
7	02
7	23
9	13
14	03

Bifiltration



10.8

7

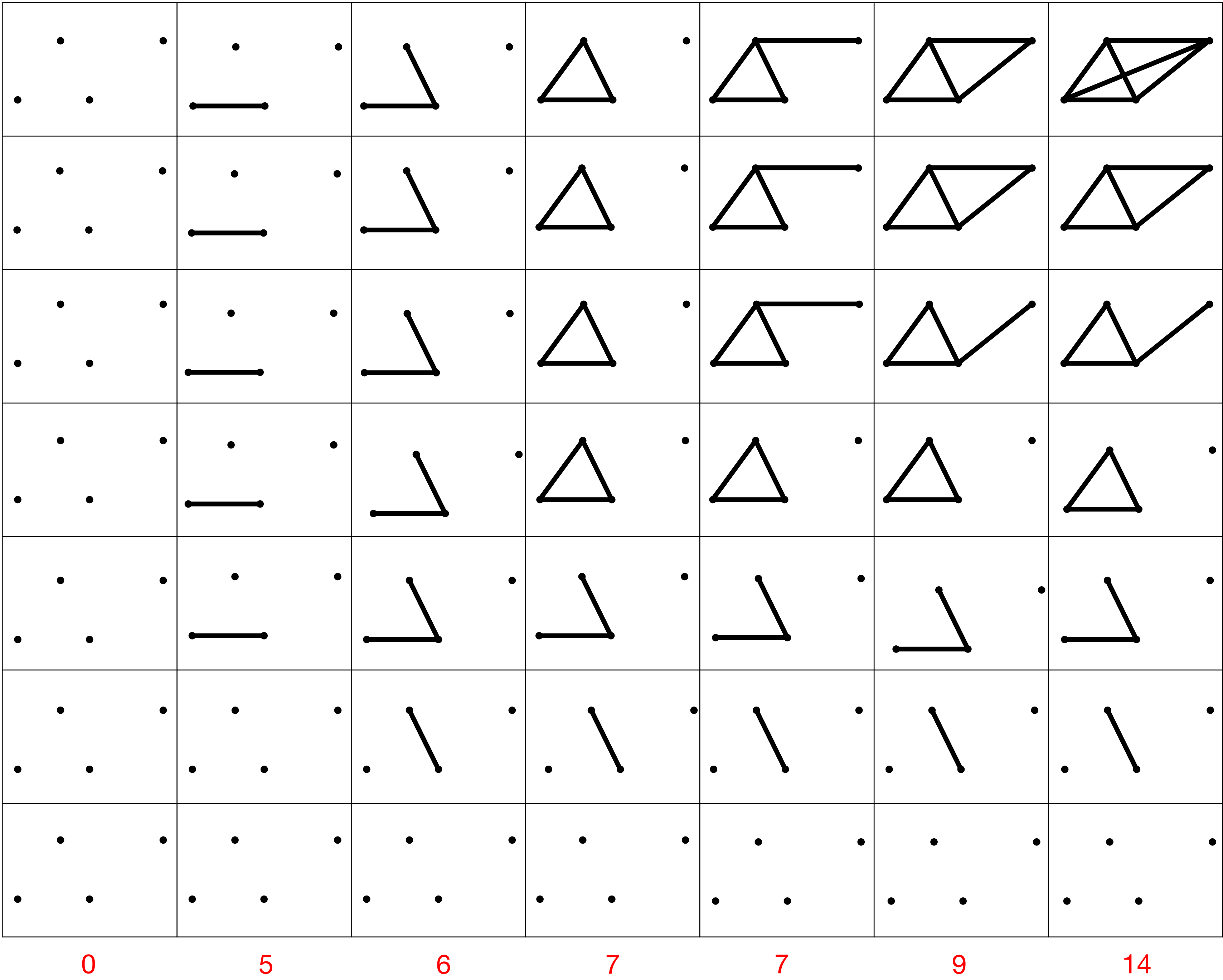
6.4

5

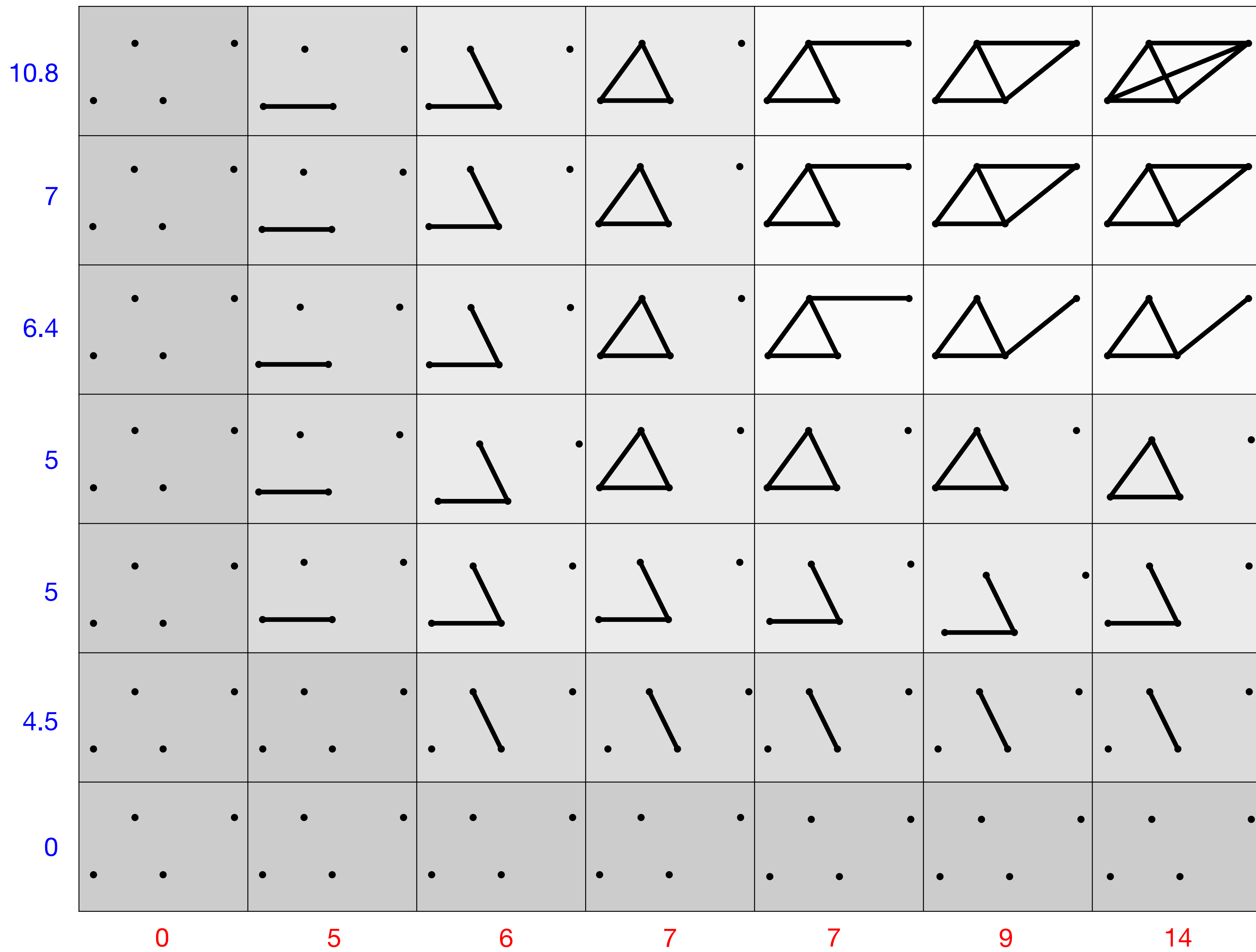
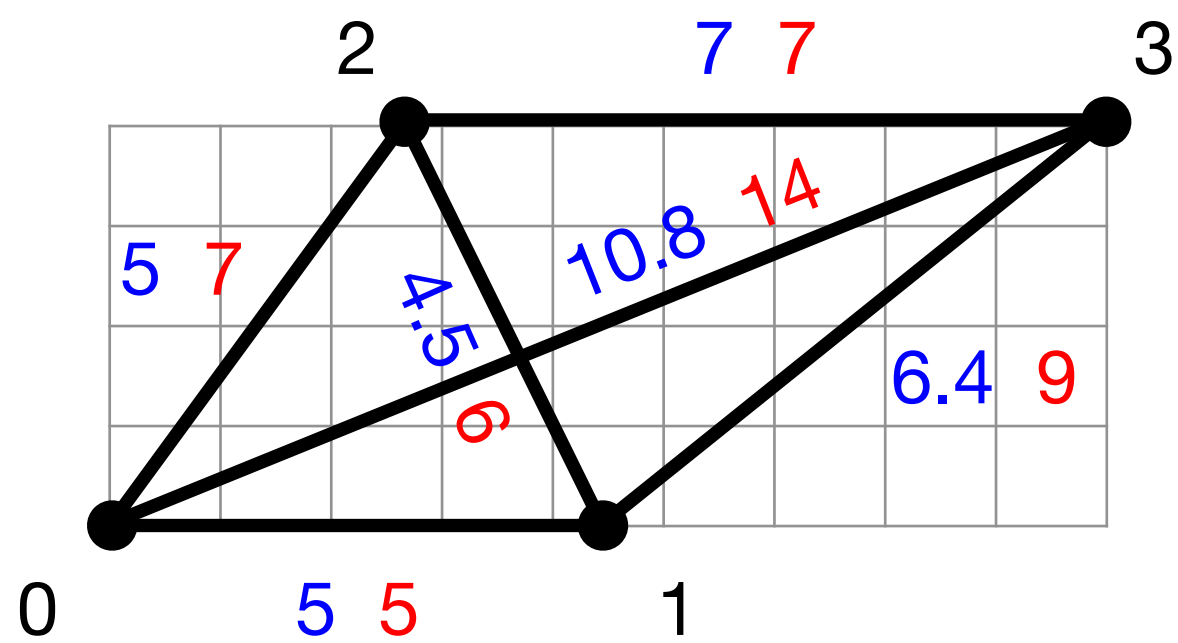
5

4.5

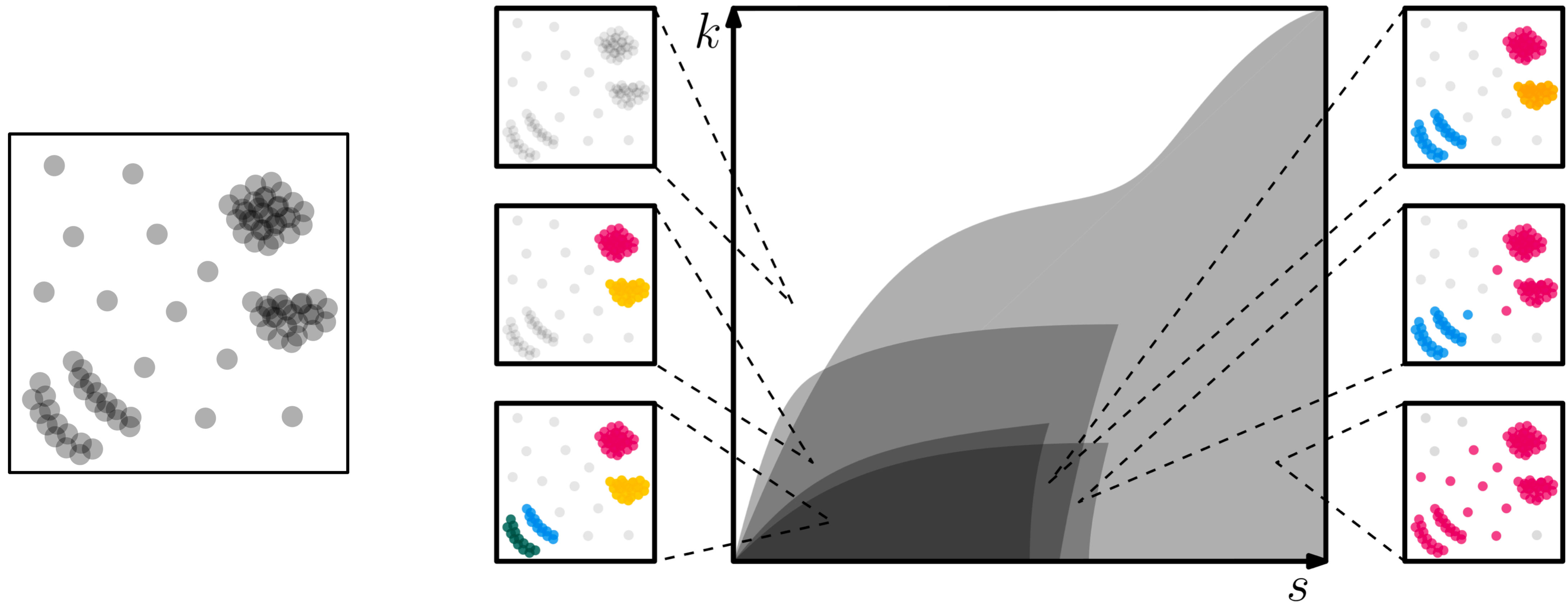
0



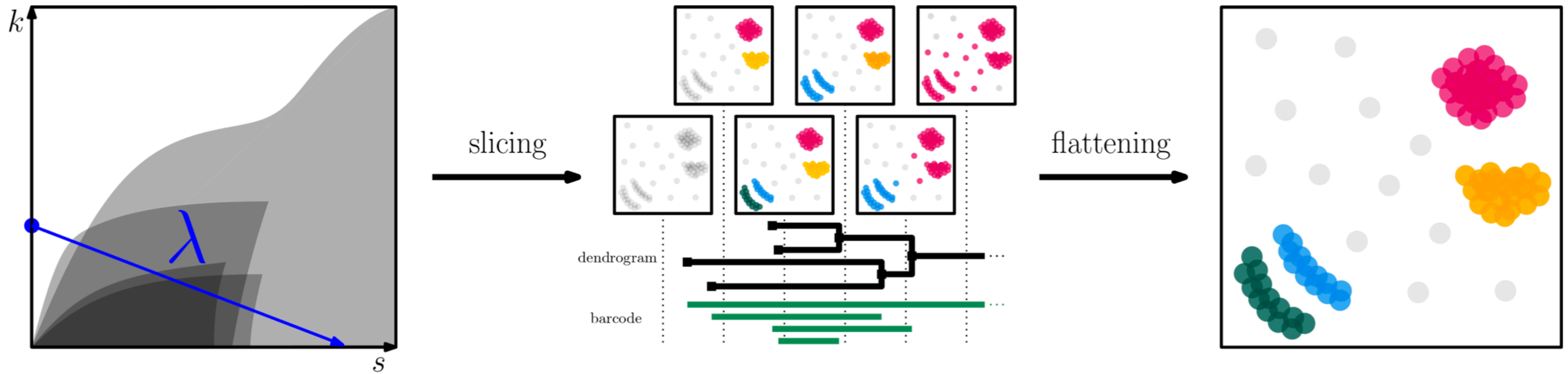
Bifiltration



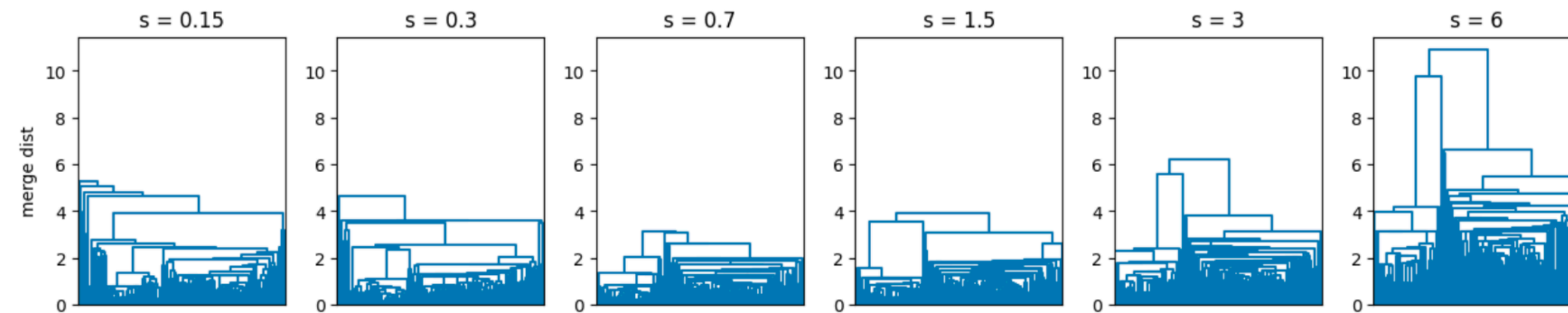
Persistable



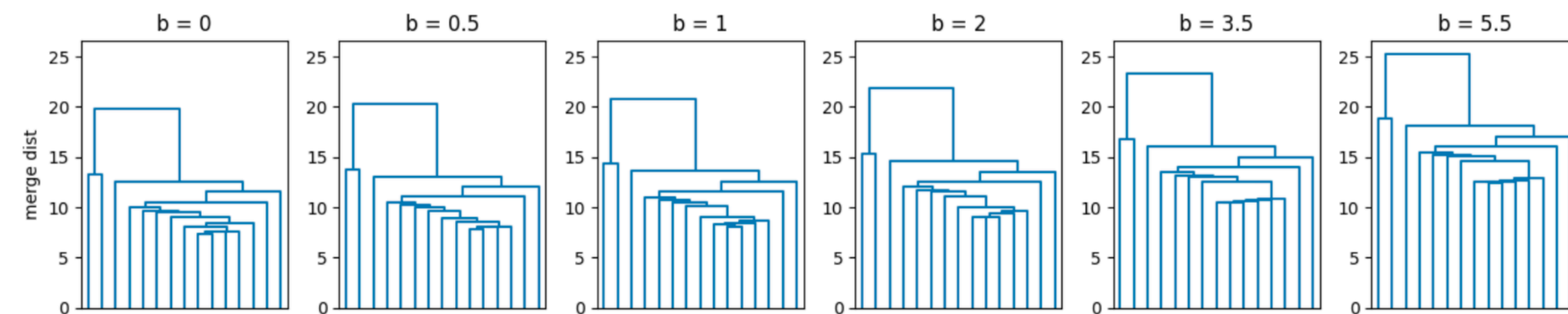
Slicing



Slicing



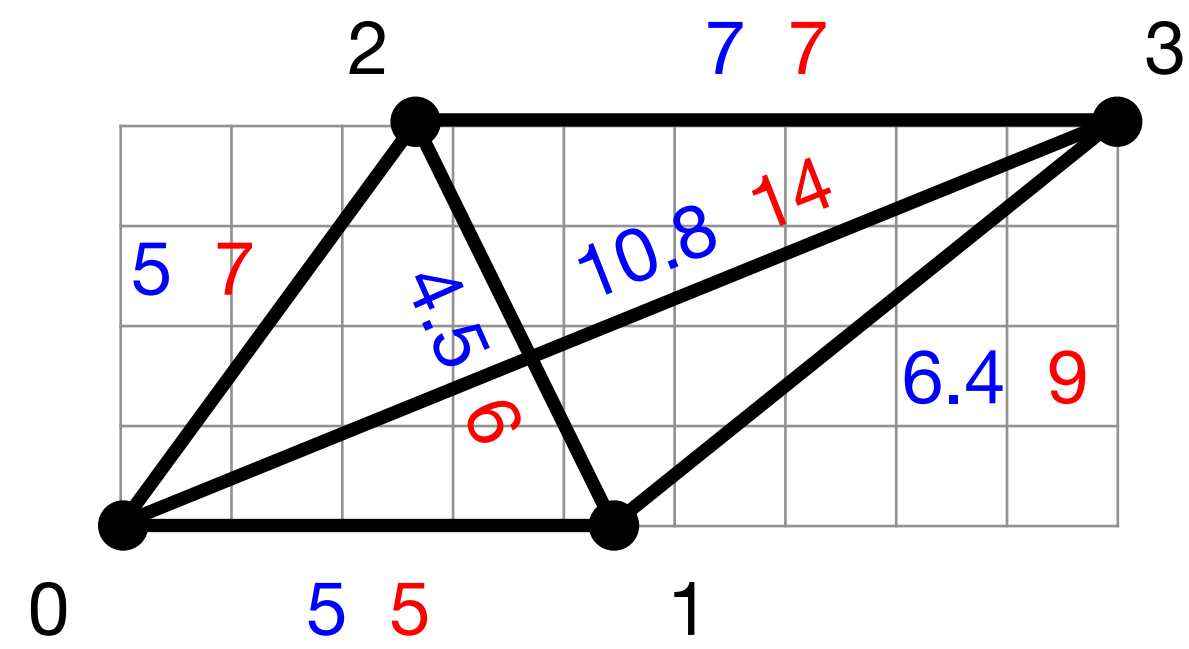
Изменение наклона



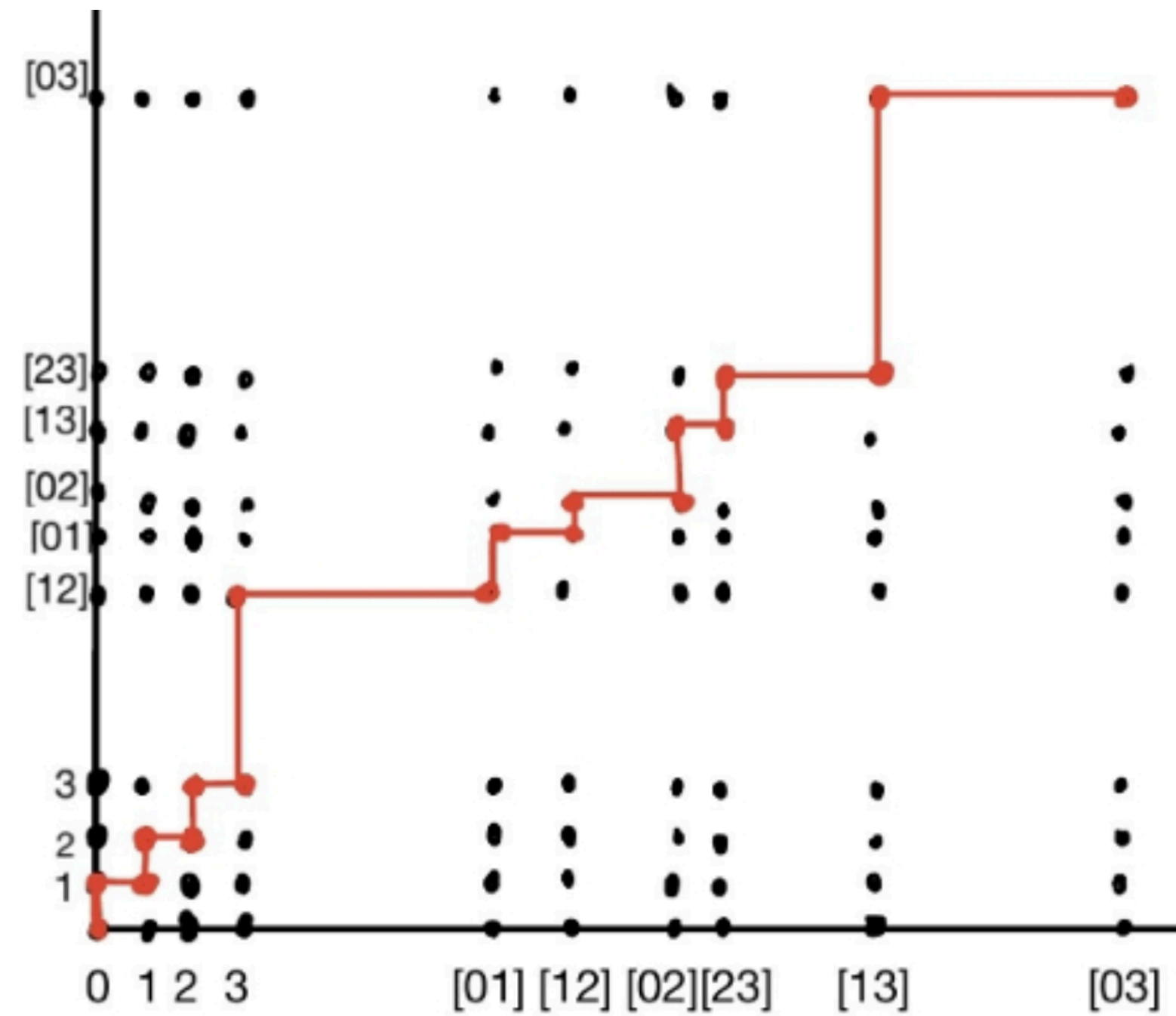
Изменение смещения

Параметры среза задают порядок появления рёбер. Наклон меняет относительный вес двух осей, а смещение меняет влияние малых значений второй оси. Поэтому при тех же данных и тех же фильтрациях получаются разные деревья слияний

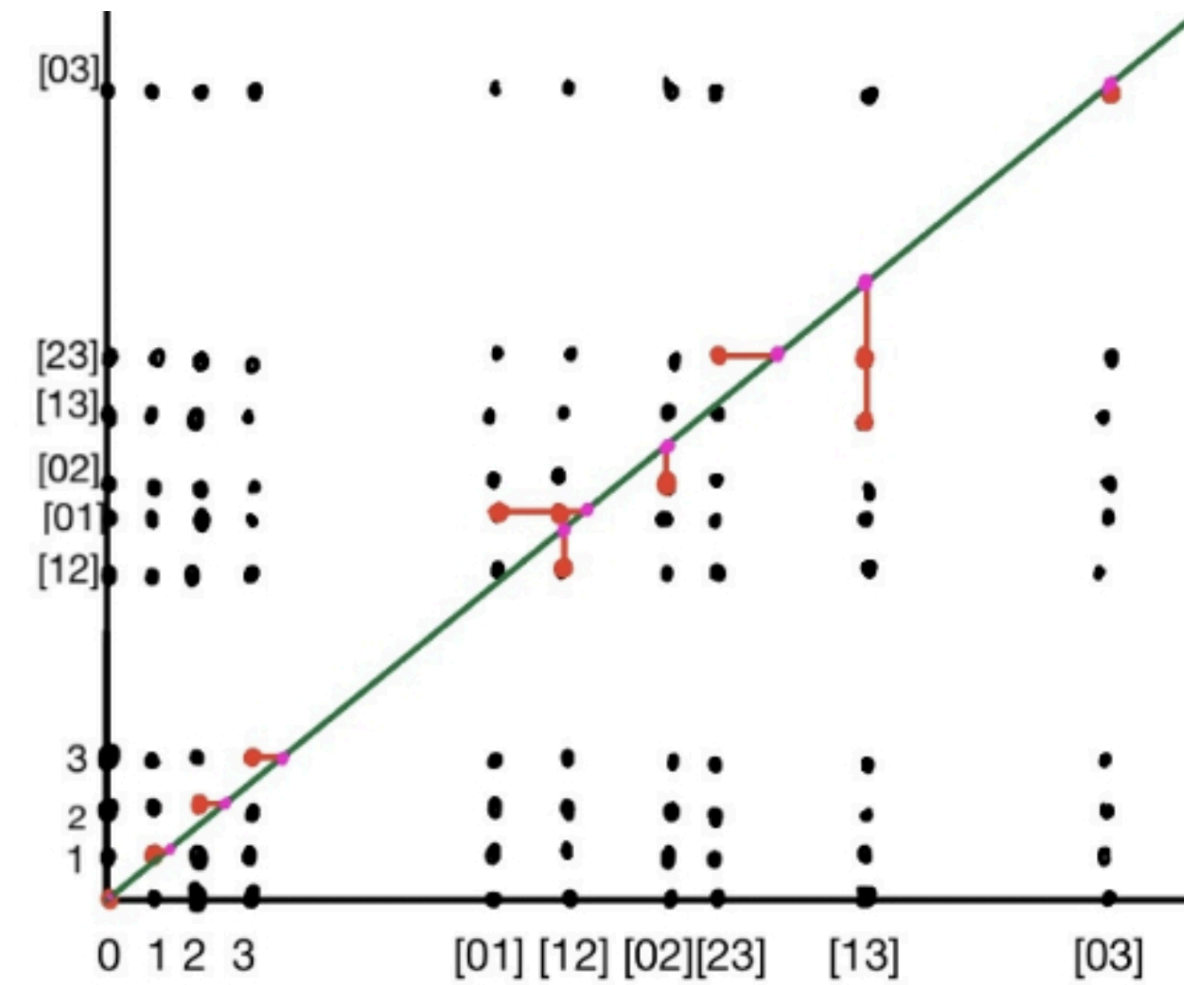
Slicing



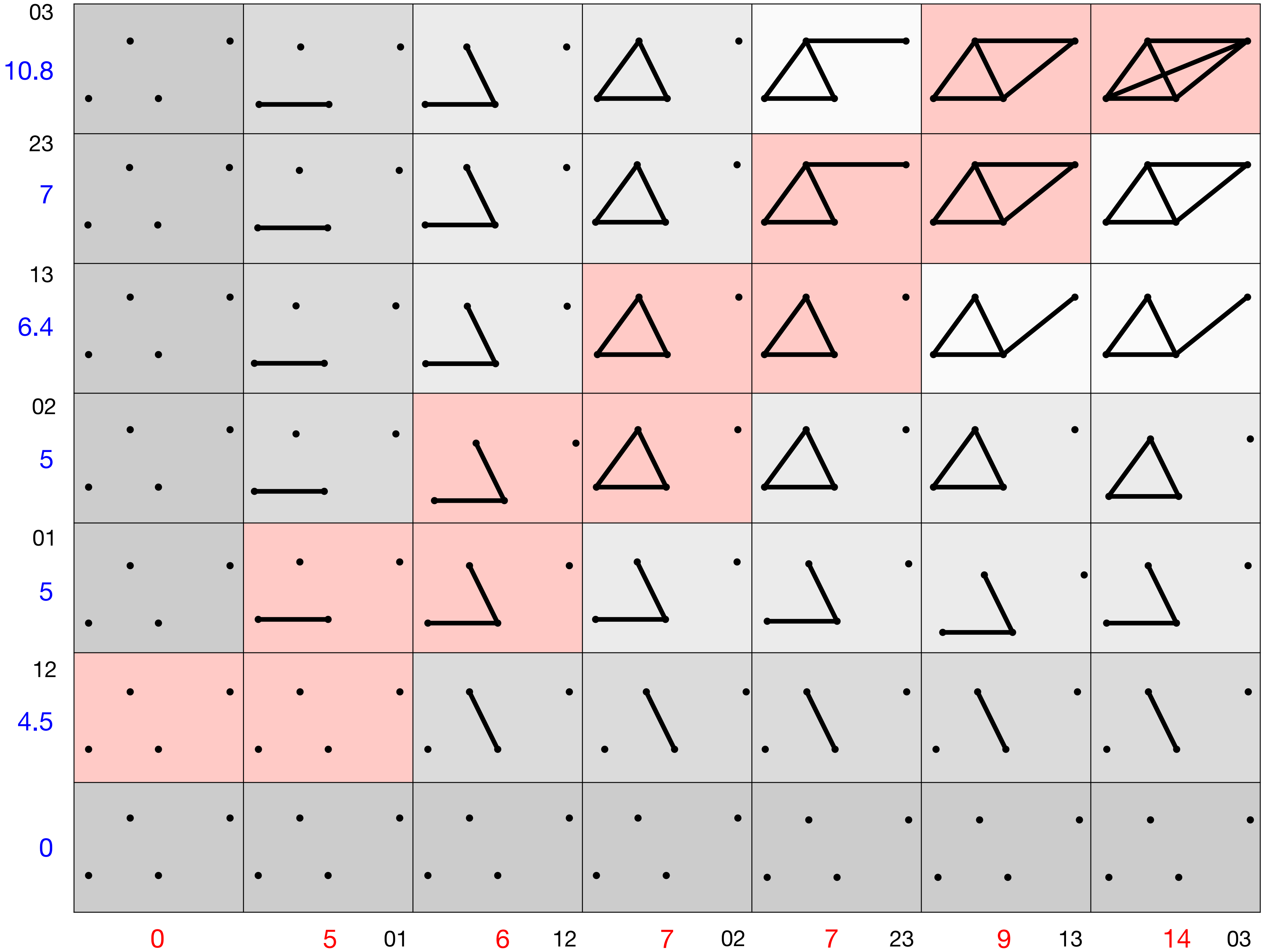
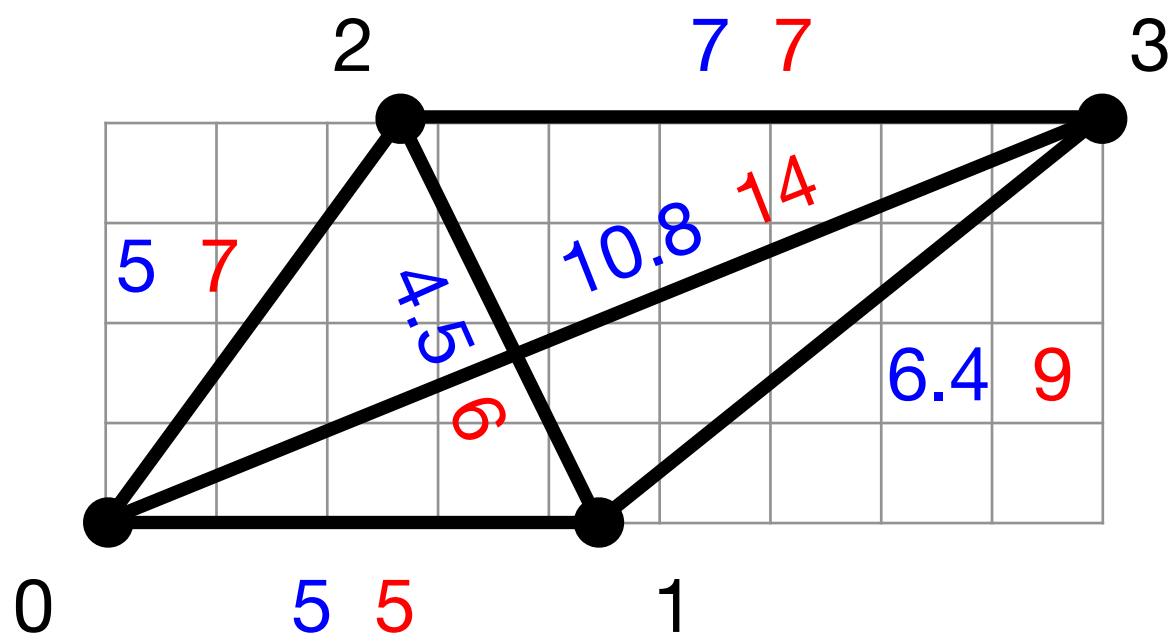
Path slice



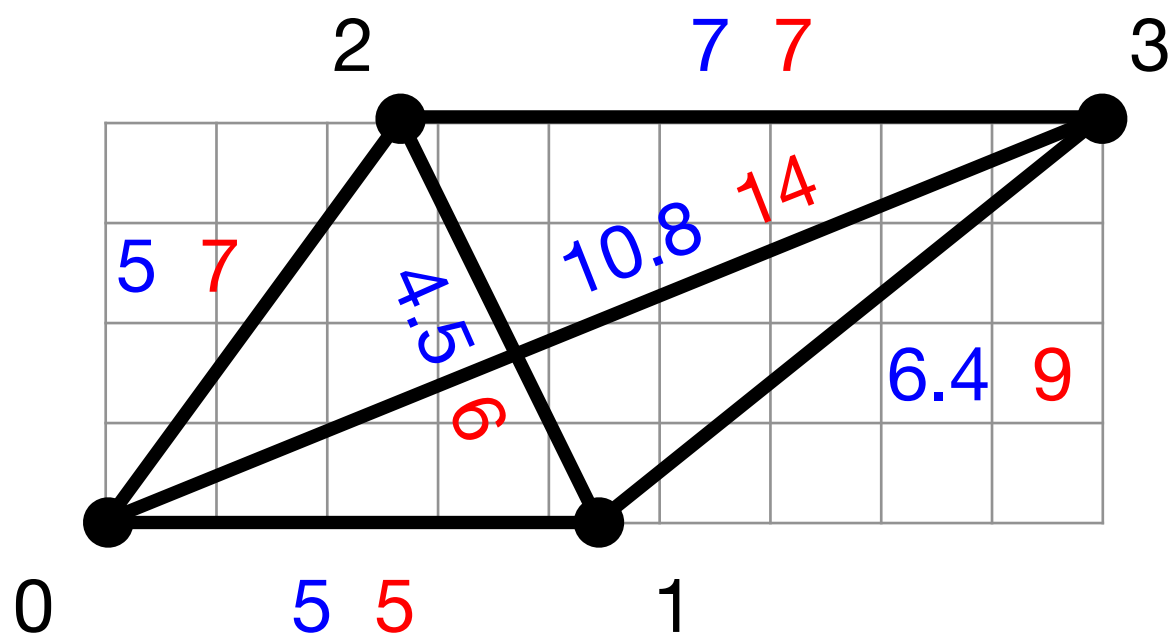
Line slice



Slicing

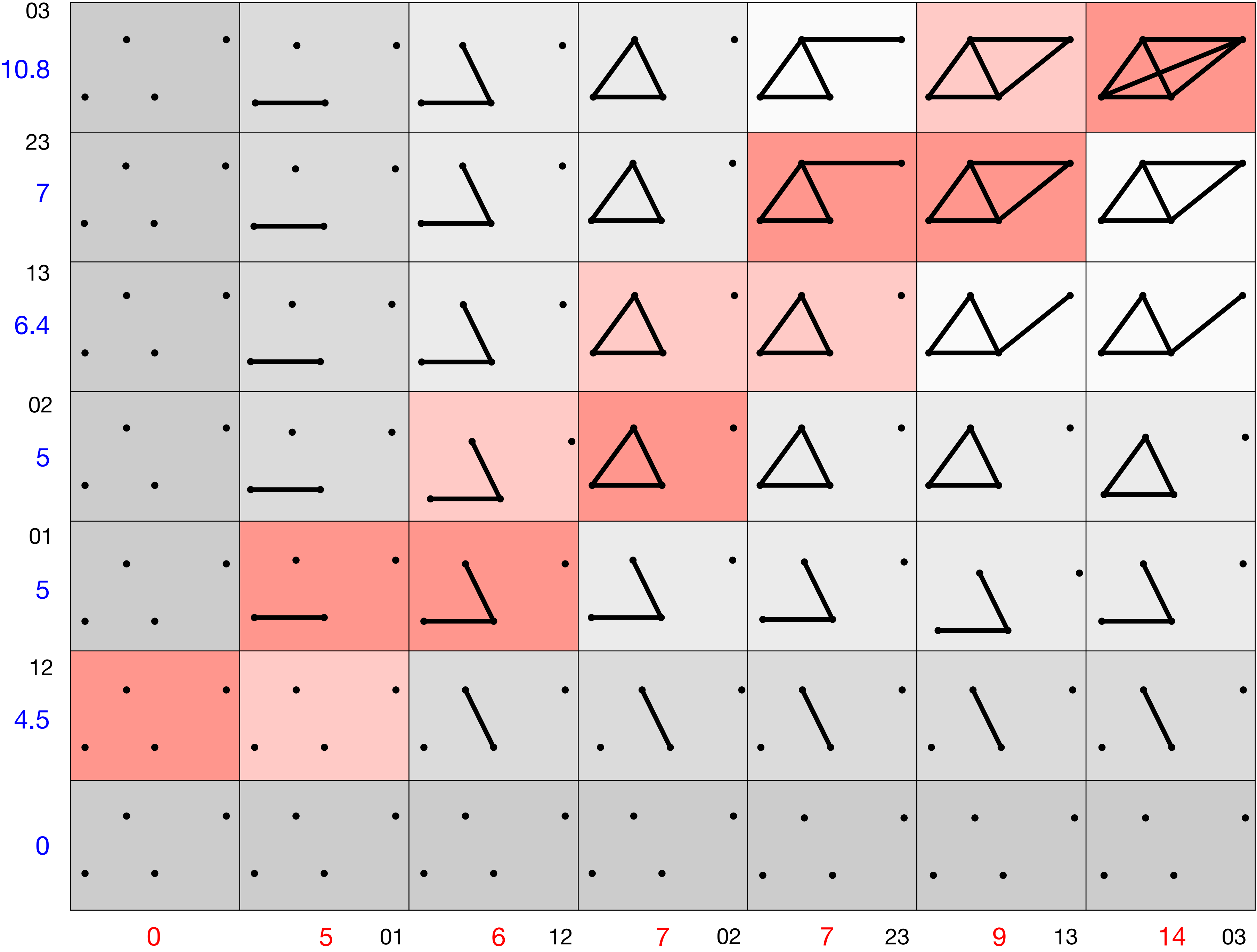


Slicing

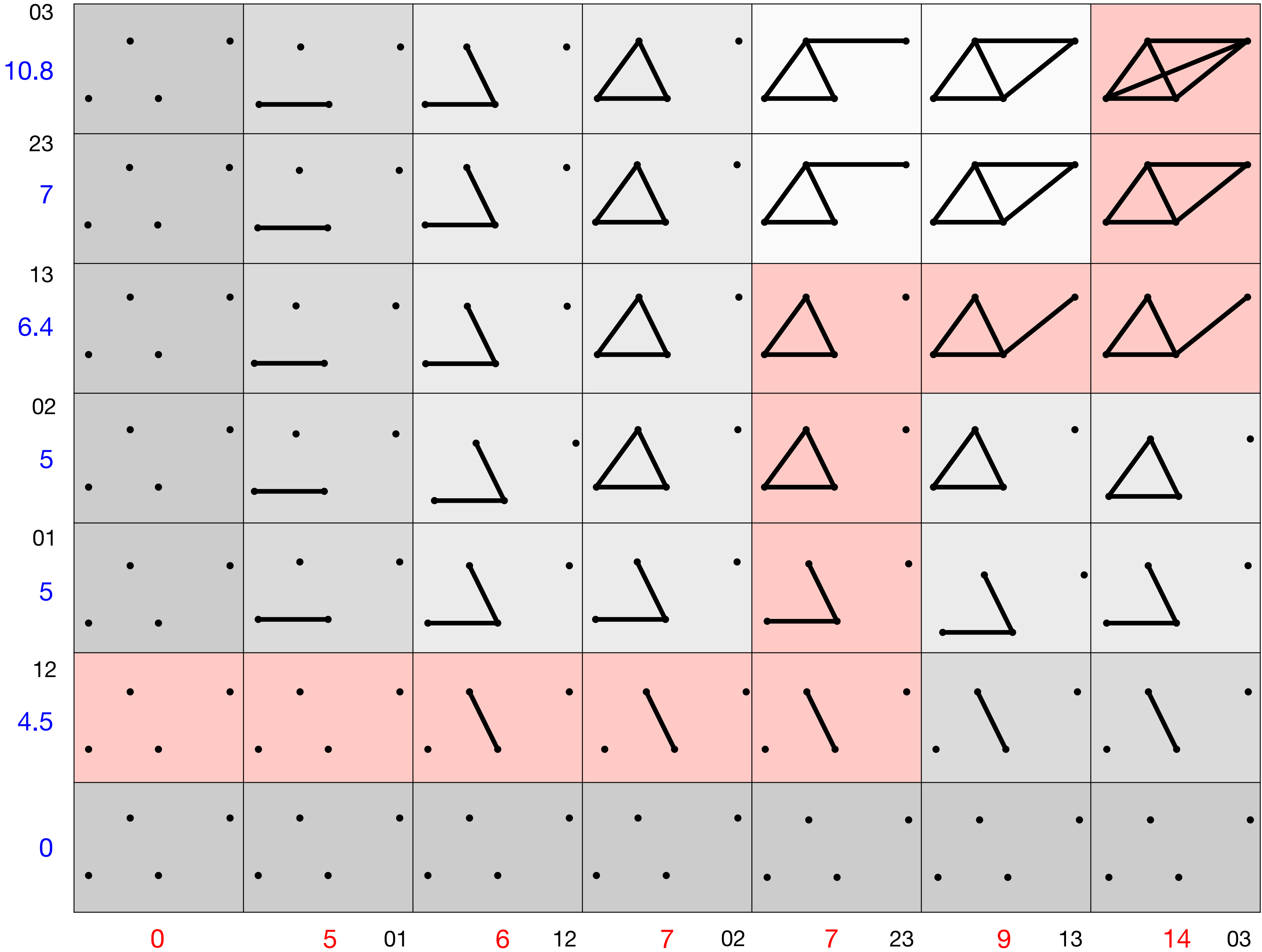
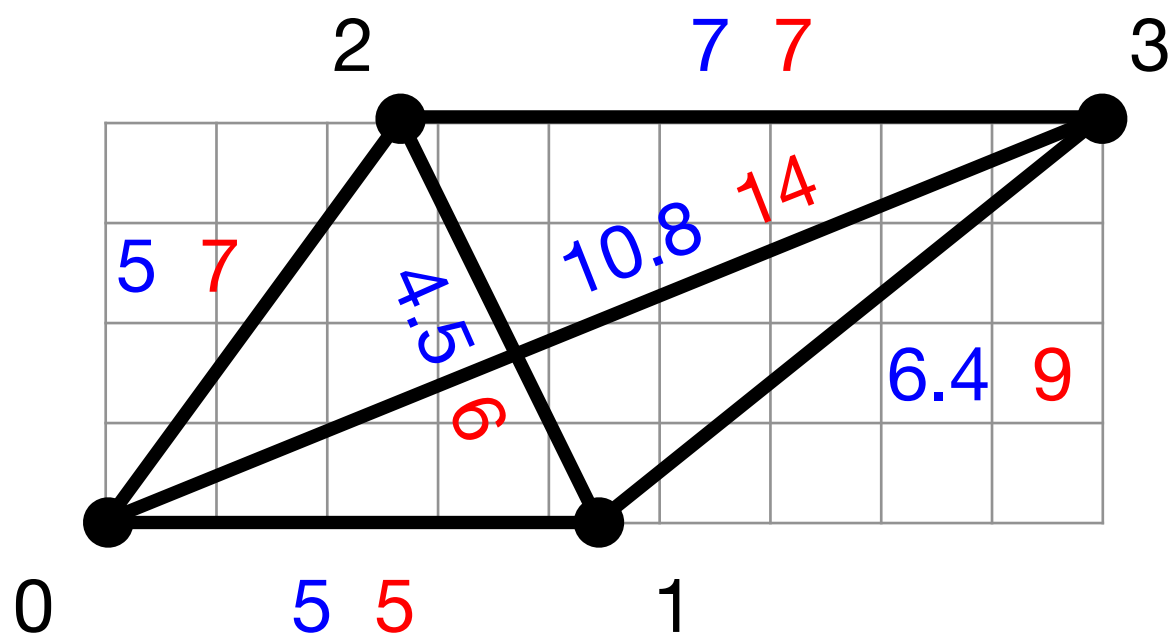


Filtration ordering

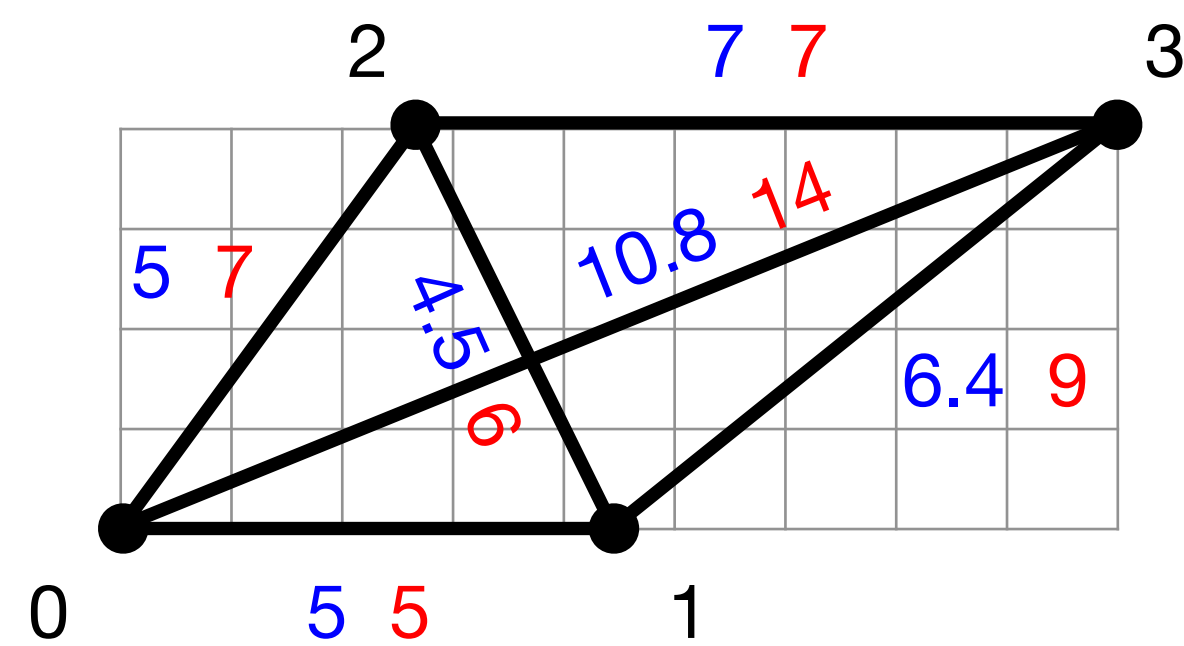
- 0, 1, 2, 3, 01, 12, 02, 23, 13, 03



Slicing

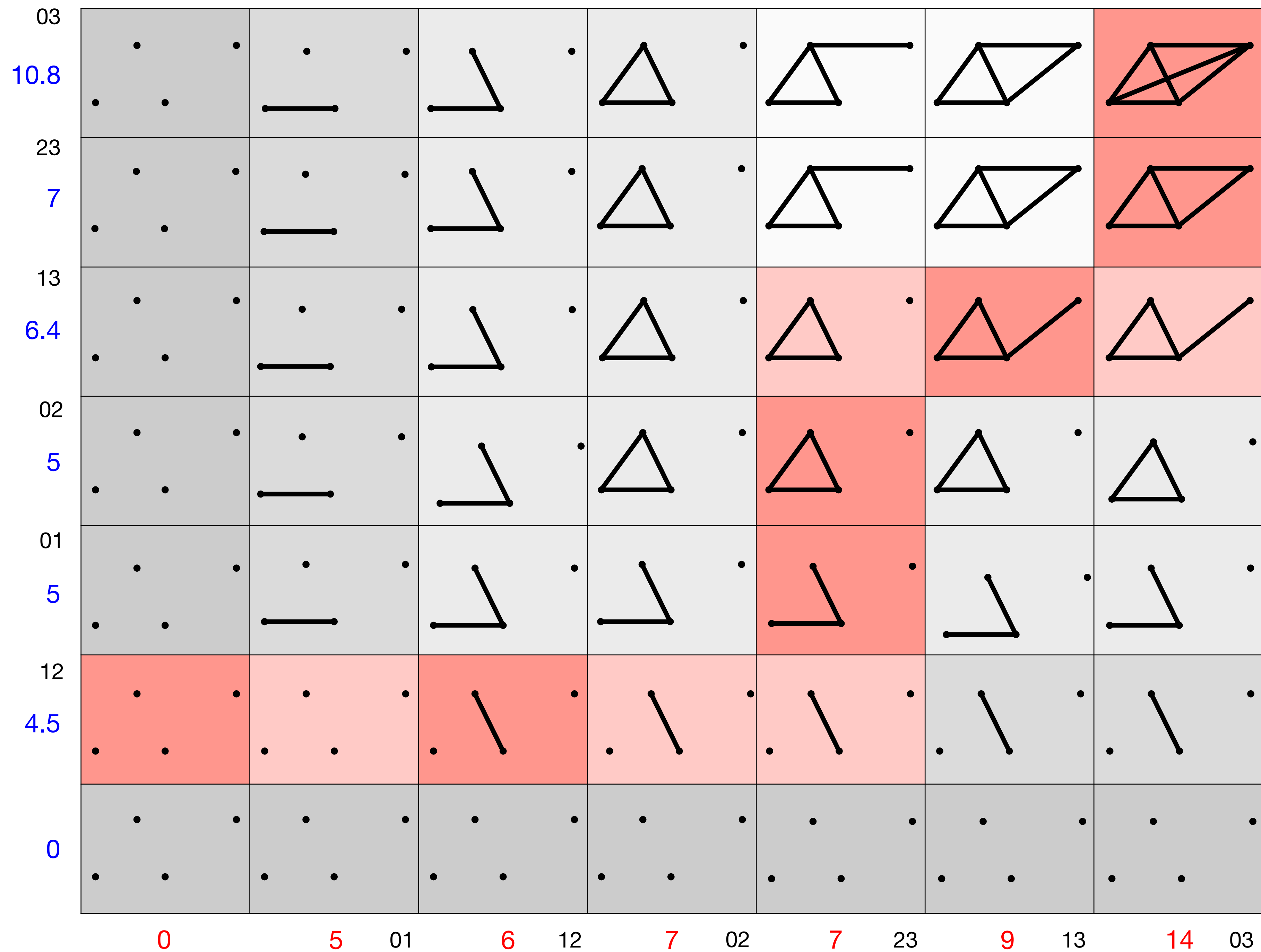


Slicing

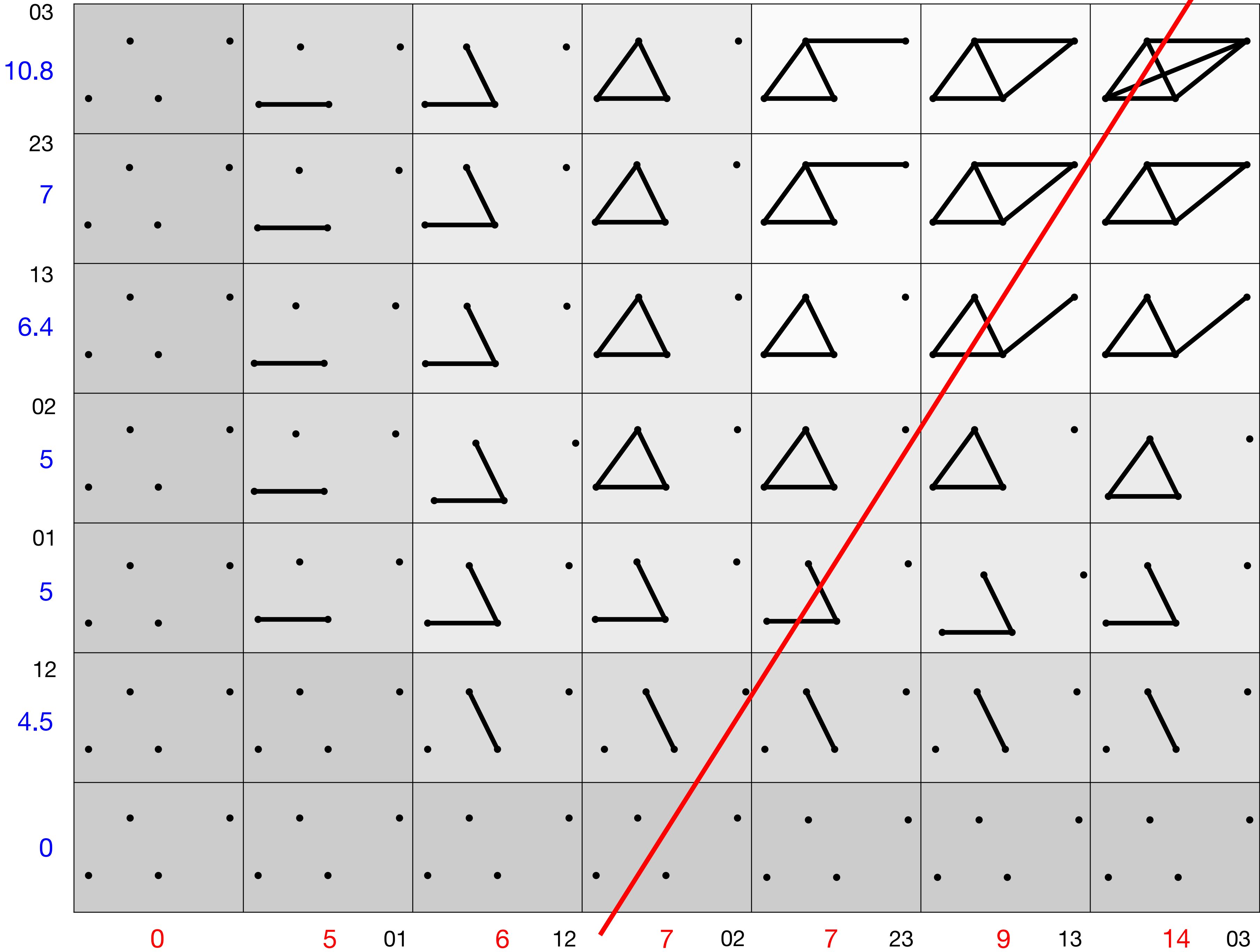
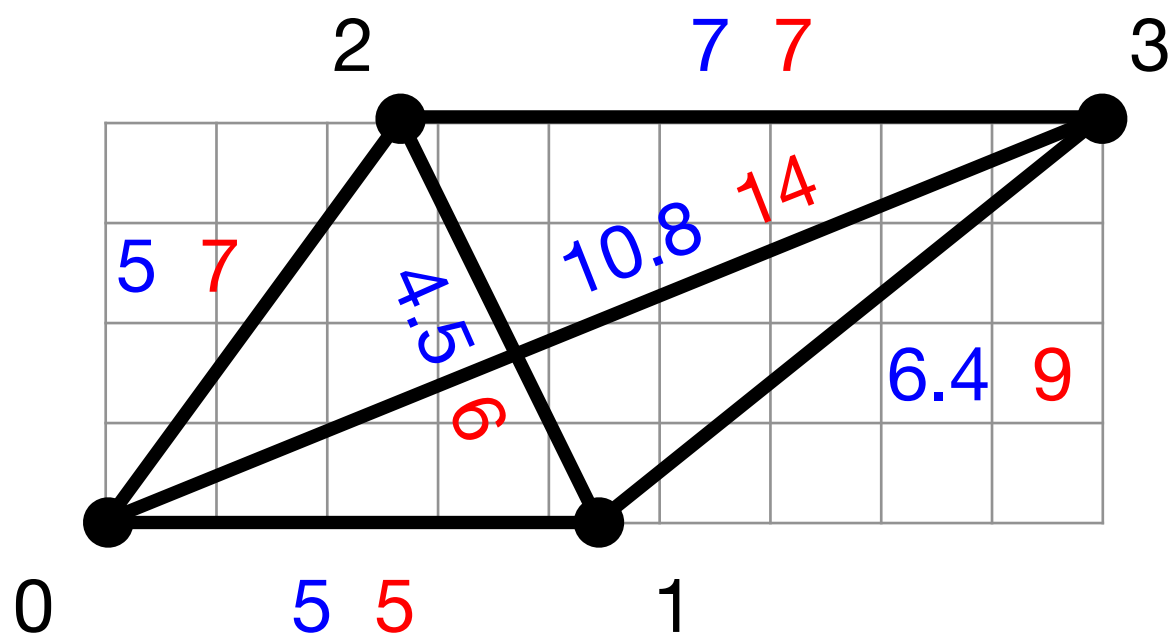


Filtration ordering

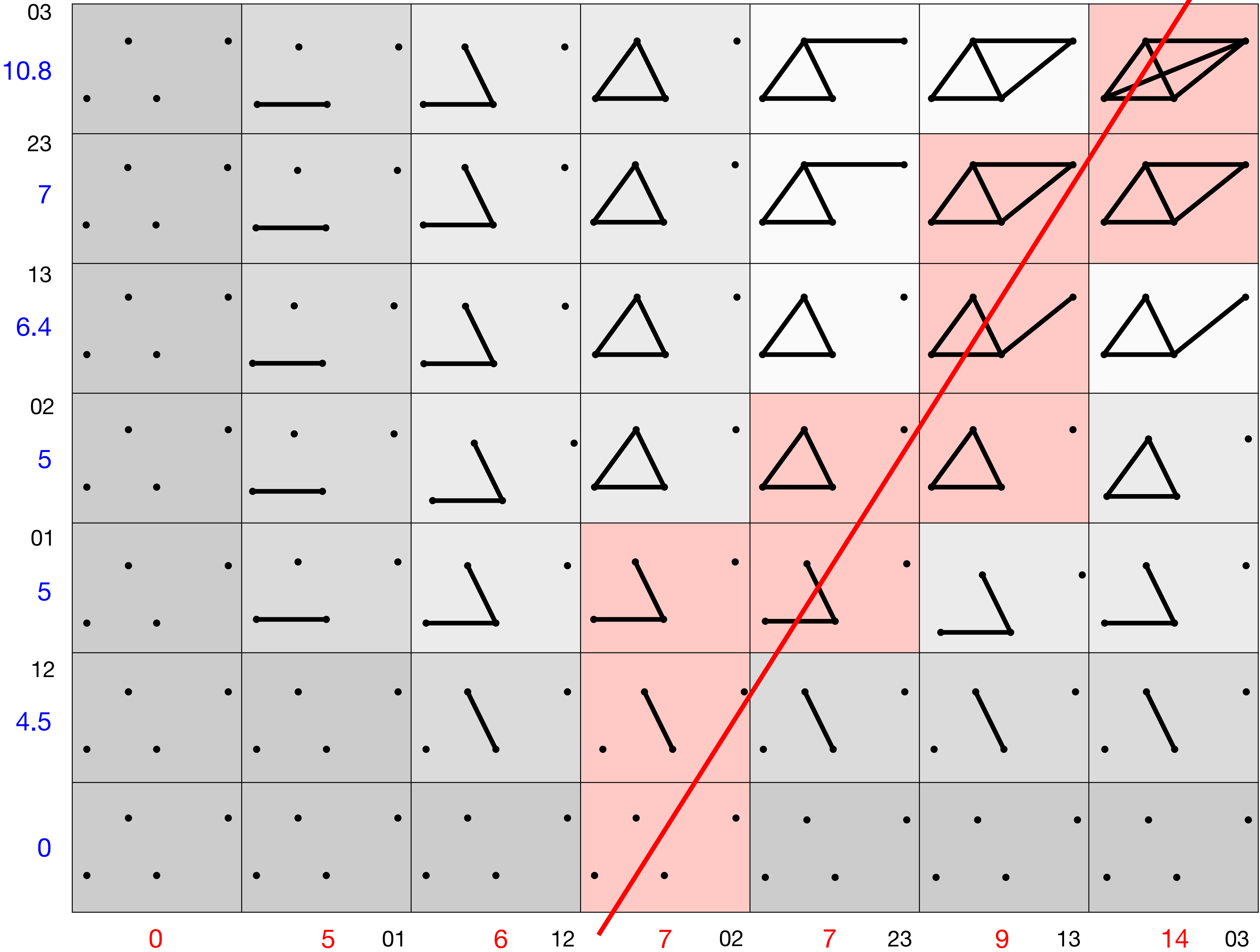
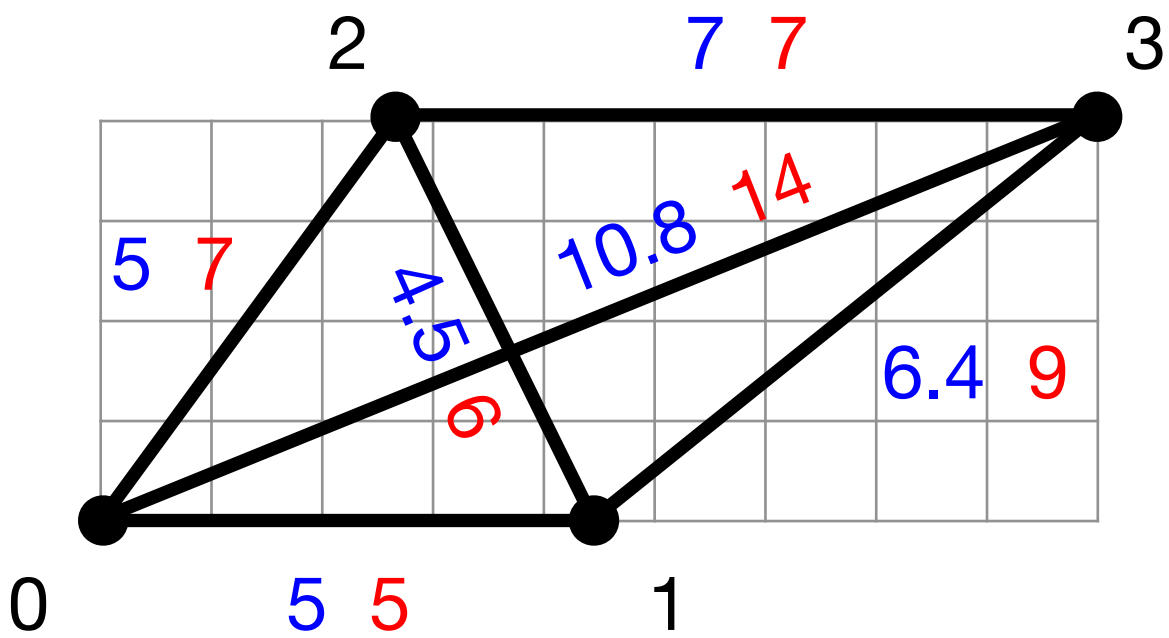
- 0, 1, 2, 3, 12, 01, 02, 12, 23, 03



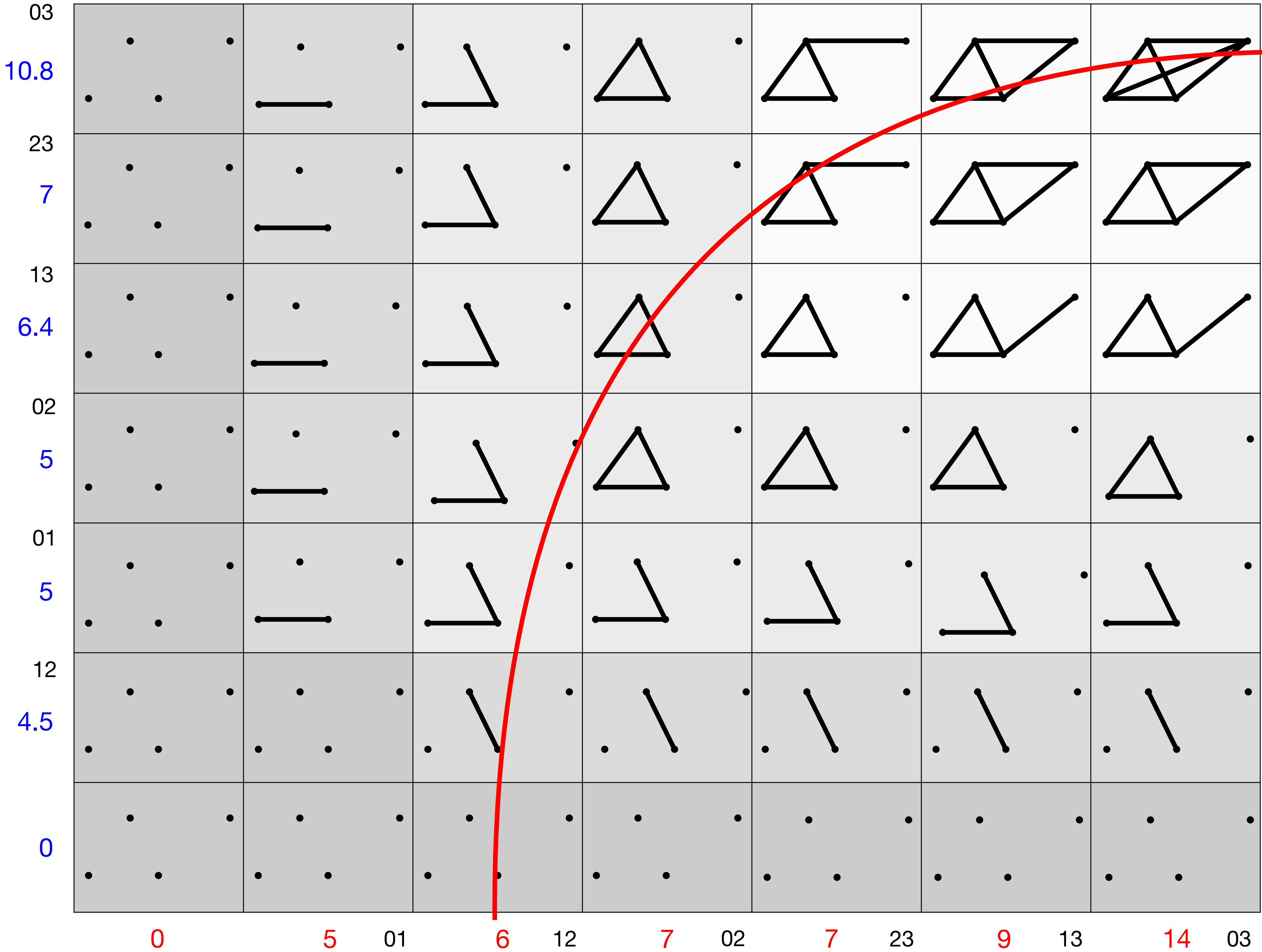
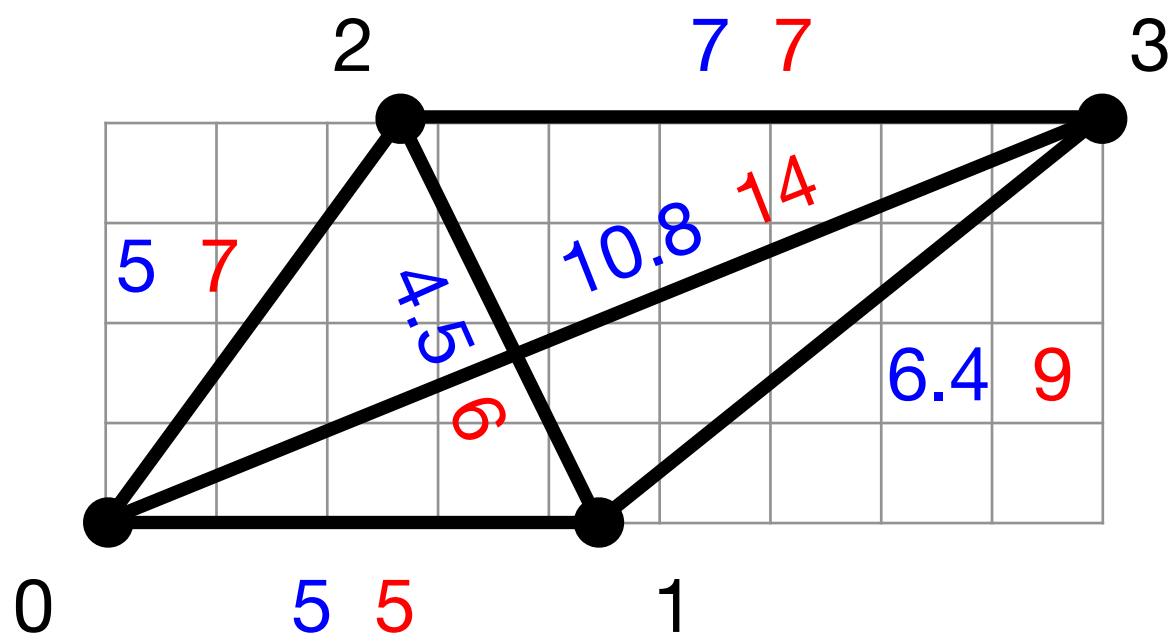
Slicing



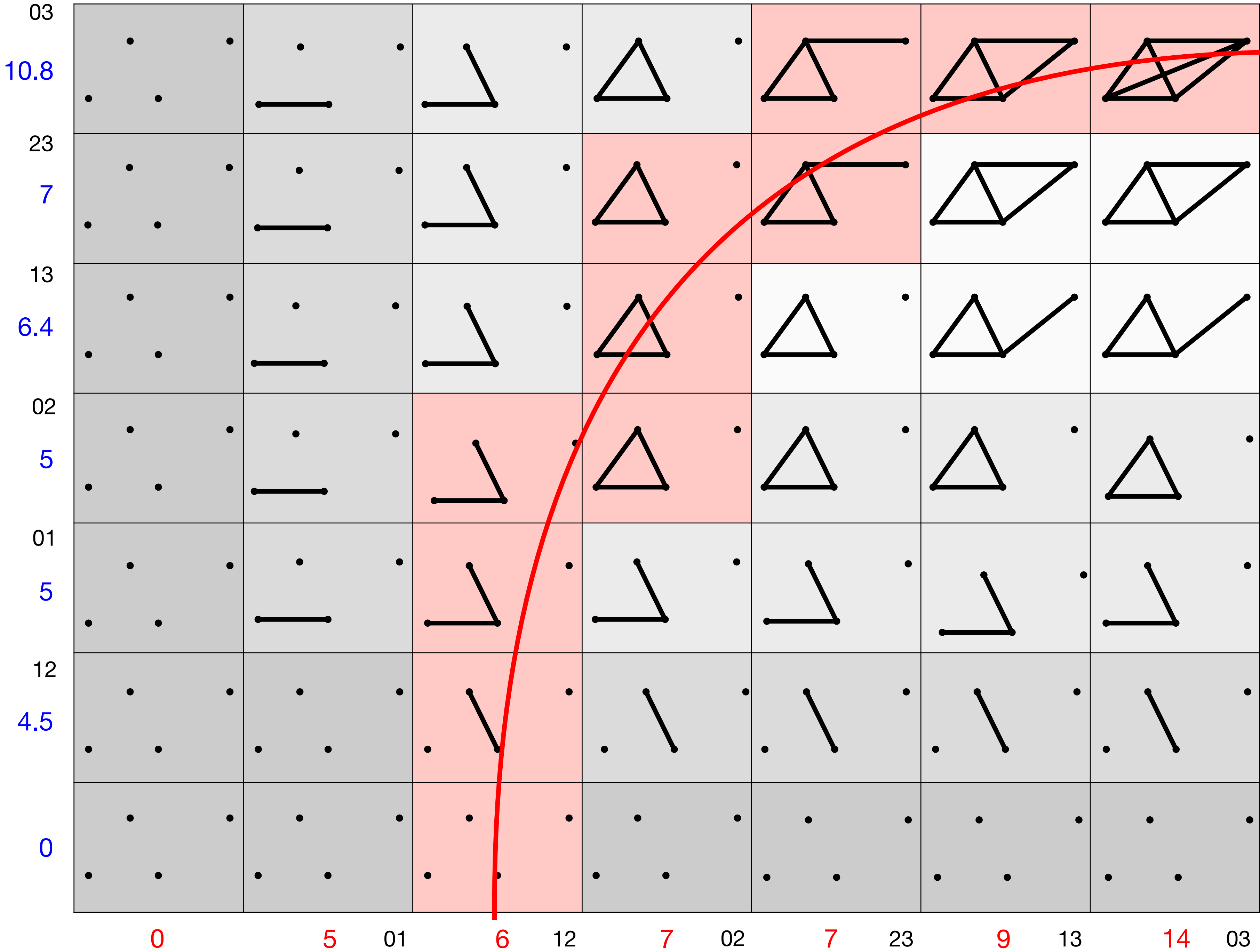
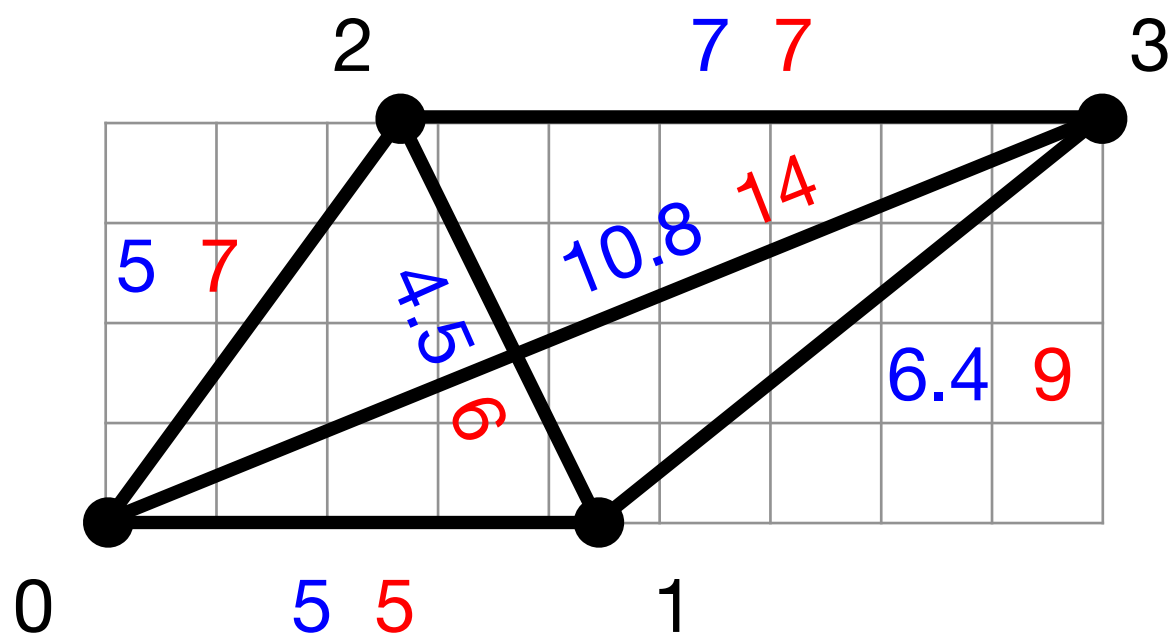
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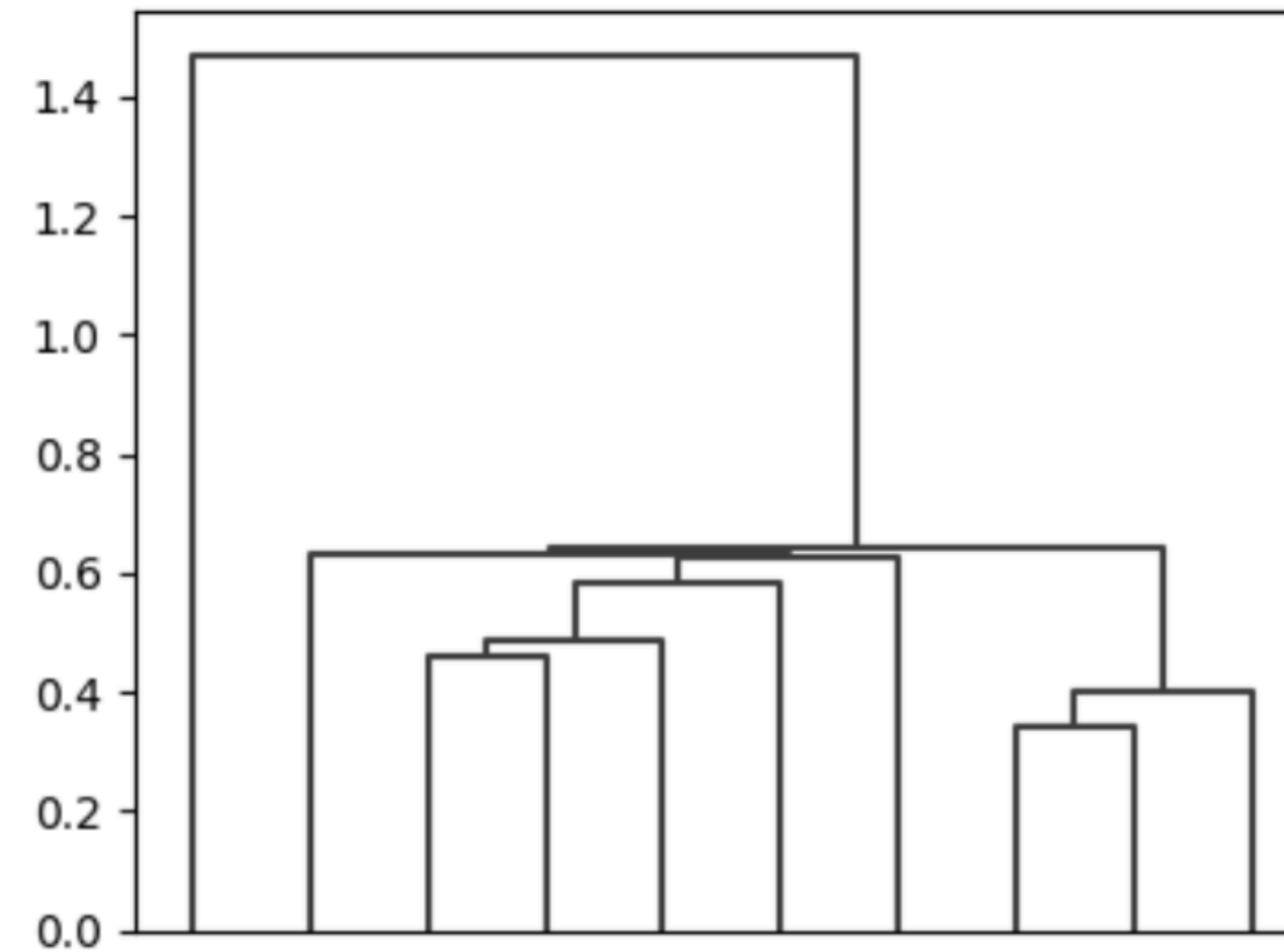
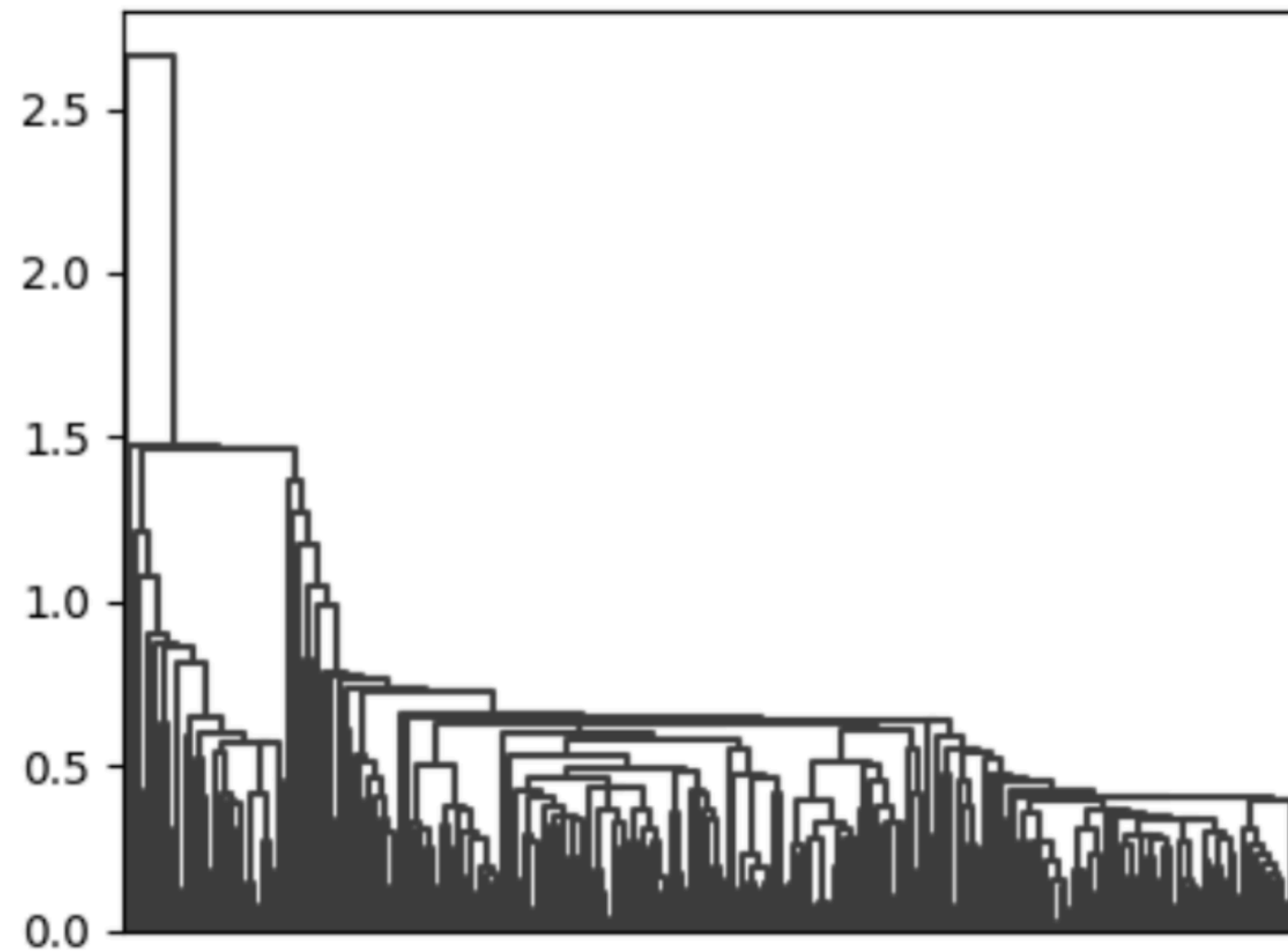
Slicing



Slicing



Simplification

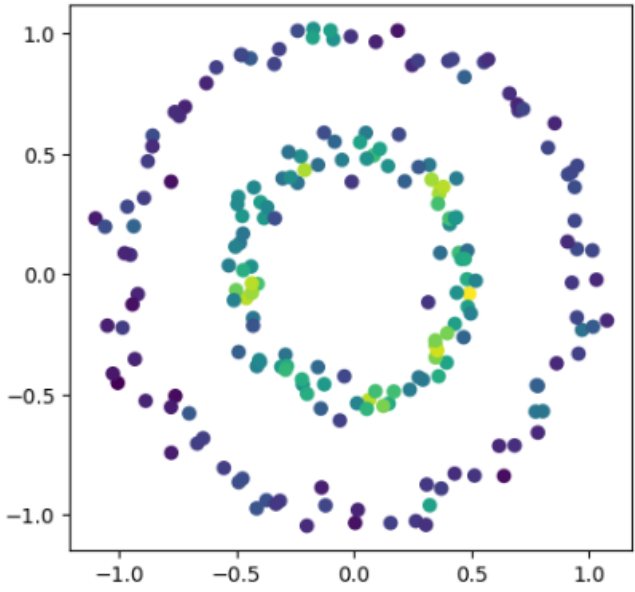


Упрощение превращает дендрограмму в более компактную. Малые ветви сливаются с другими компонентами, становится меньше шума

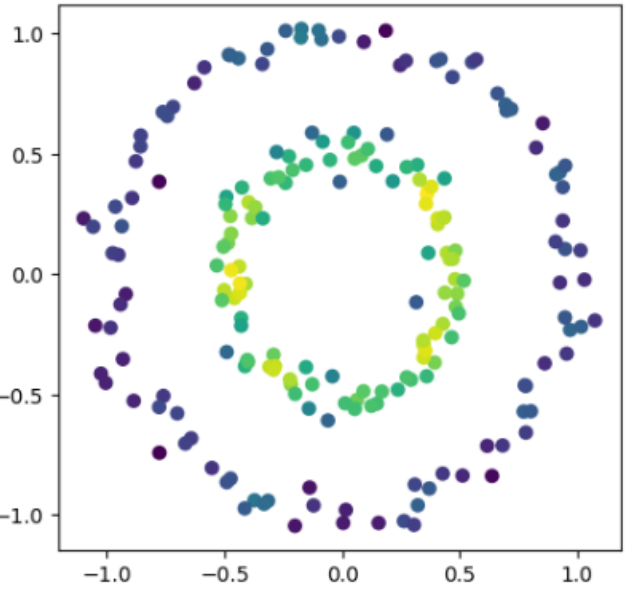
Multiparameter hierarchical clustering

Density

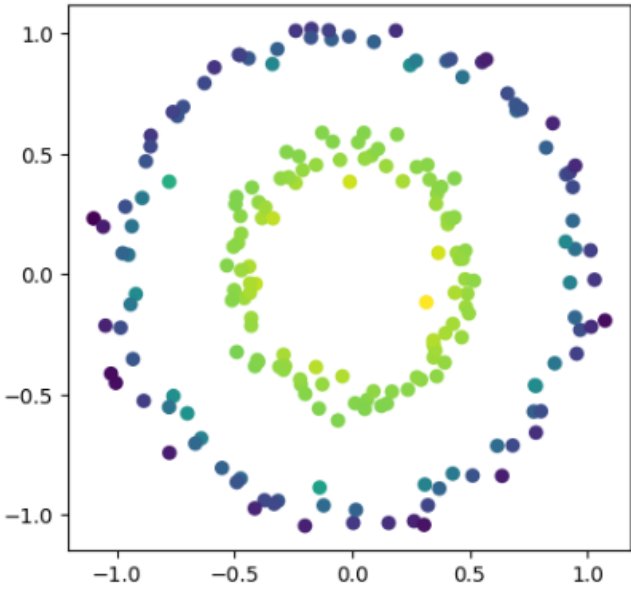
Density estimates



kNN



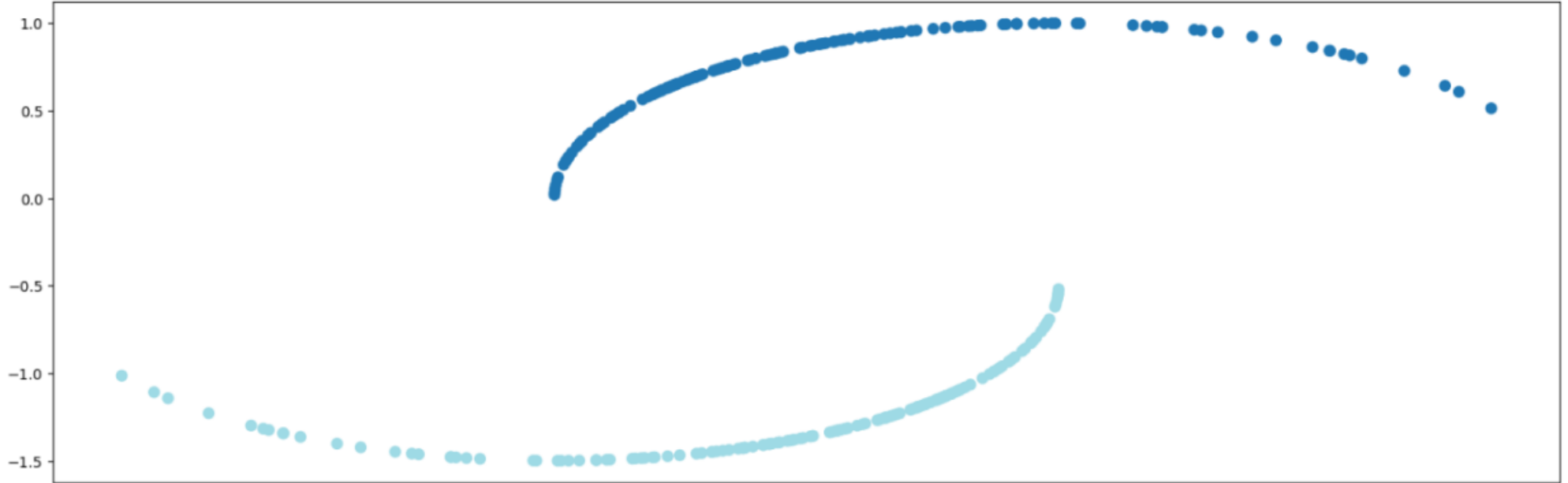
kNN-AVG



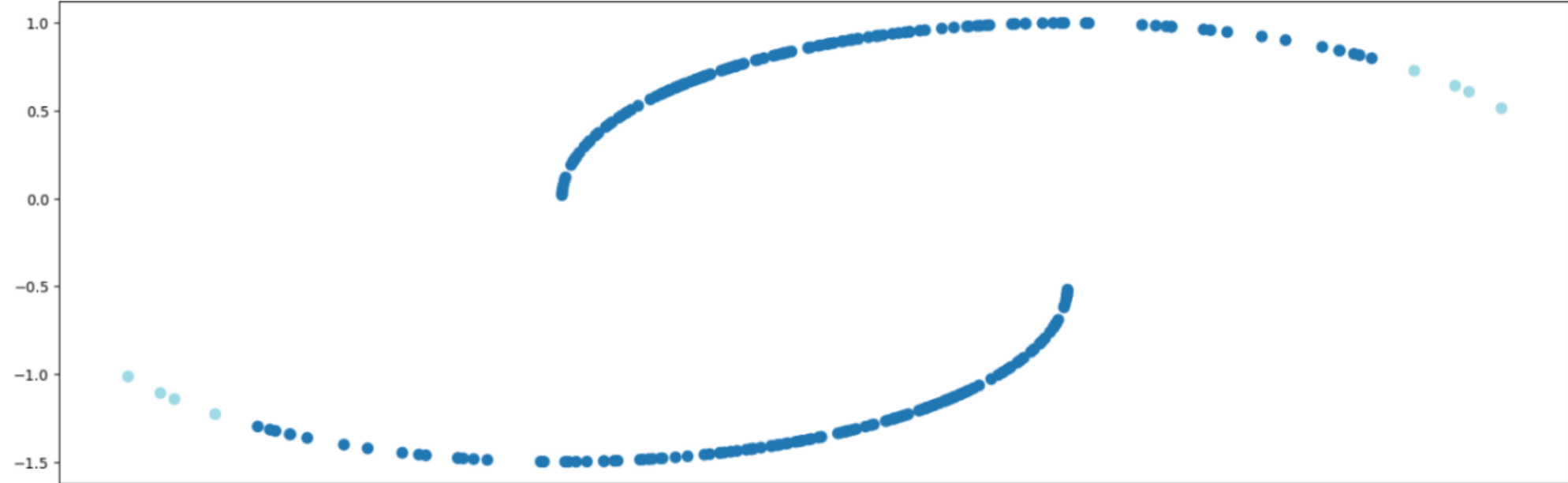
Christoffel

Методы сравнения	Pearson	Spearman
KDE vs KNN	0.922	0.873
KDE vs KNN-AVG	0.884	0.874
KDE vs Christoffel	0.992	0.891
KNN vs KNN-AVG	0.933	0.893
KNN vs Christoffel	0.898	0.806
KNN-AVG vs Christoffel	0.851	0.793

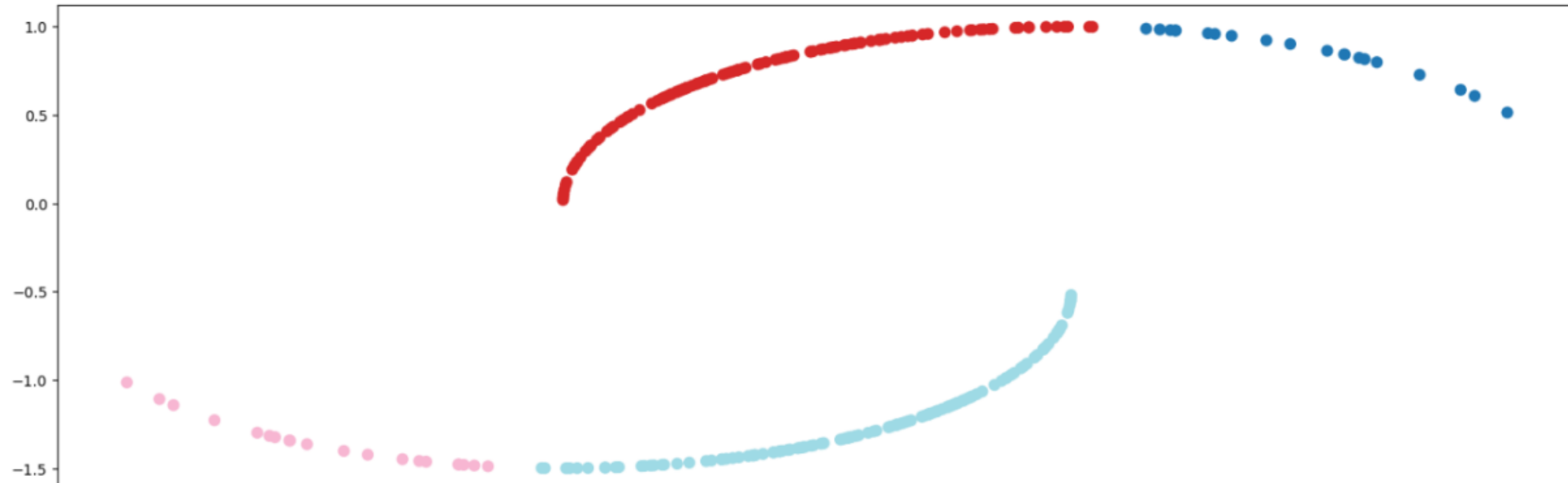
Distance



Density



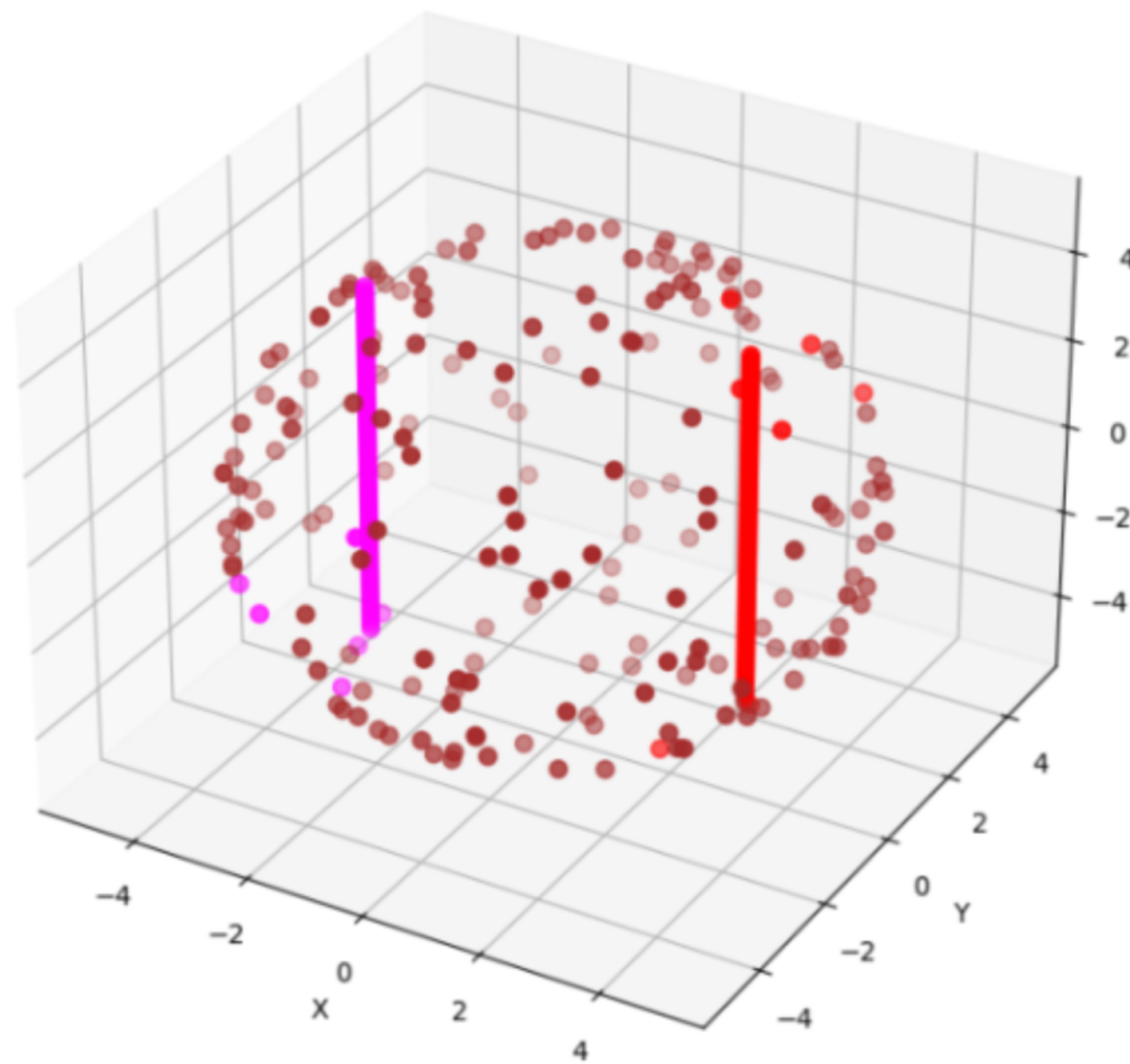
Distance + density



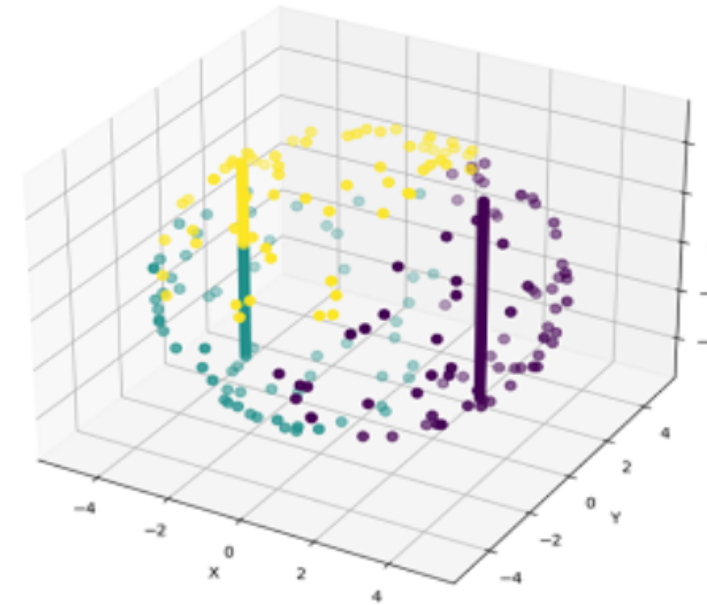
Multiparameter hierarchical clustering

Intrinsic dimension

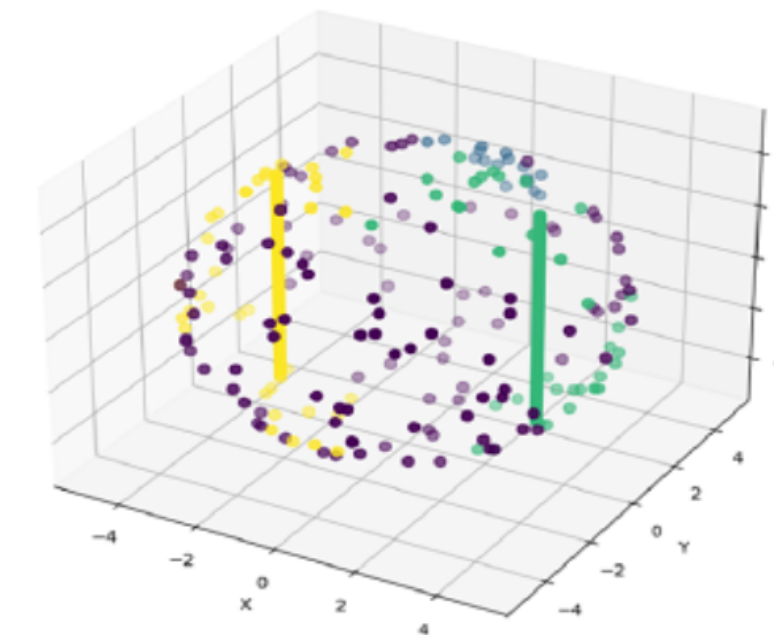
Distance + intrinsic dimension
Multiparametric



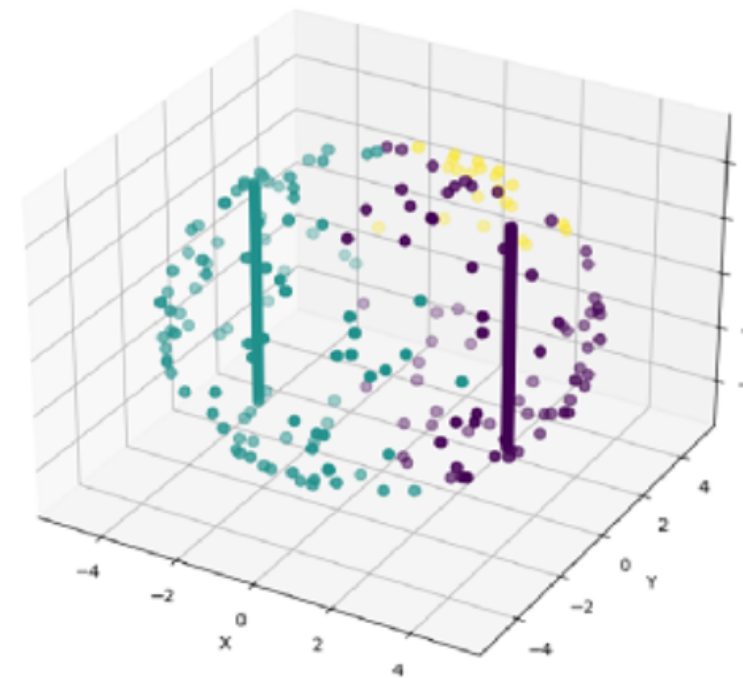
KMeans



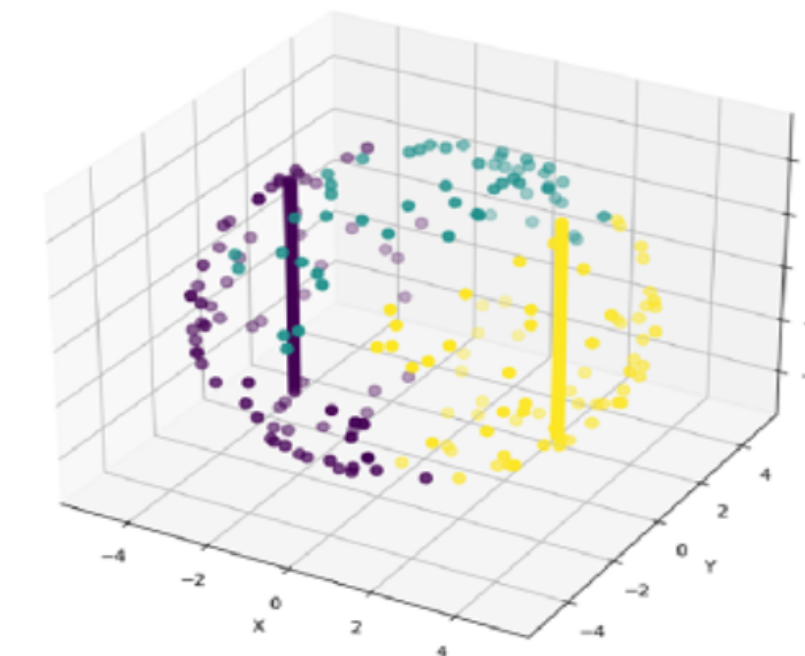
HDBSCAN



Tomato



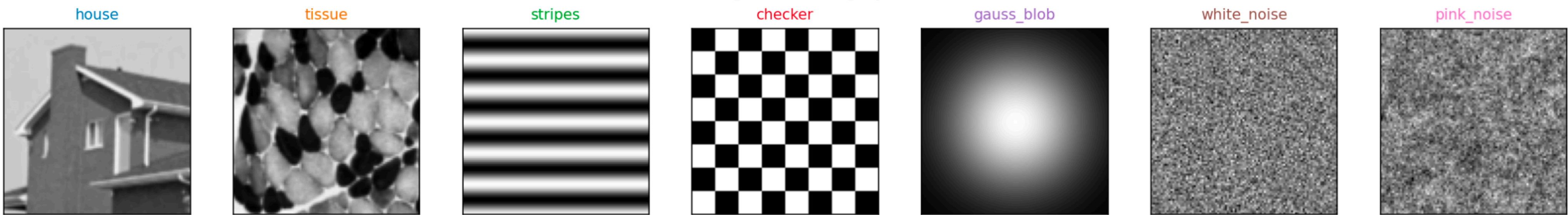
Genie



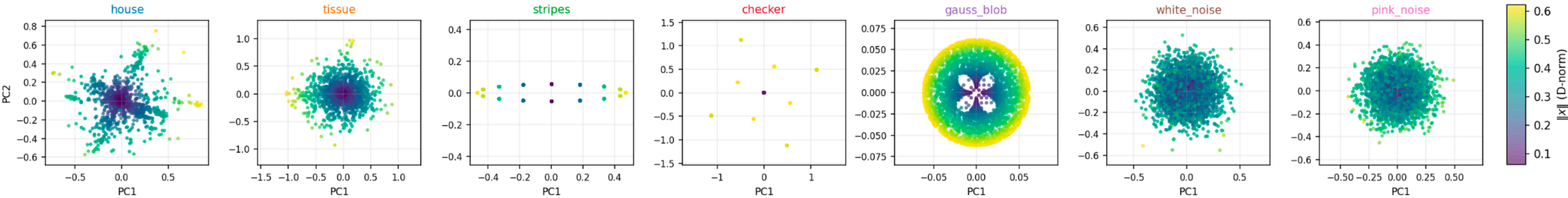
Multiparameter hierarchical clustering

Intrinsic dimension

Source images (96×96, grayscale)

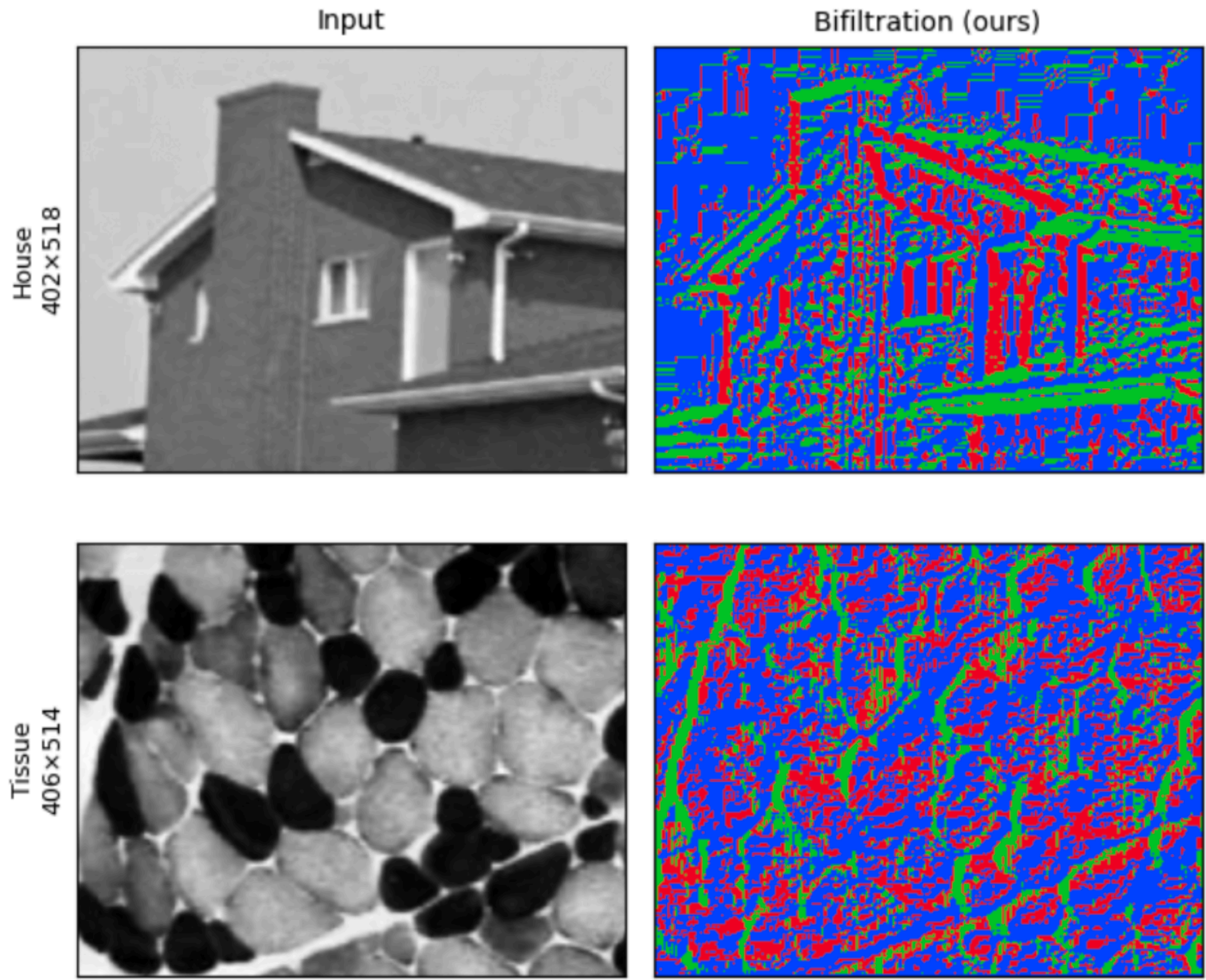


PC1 vs PC2 — raw patch cloud

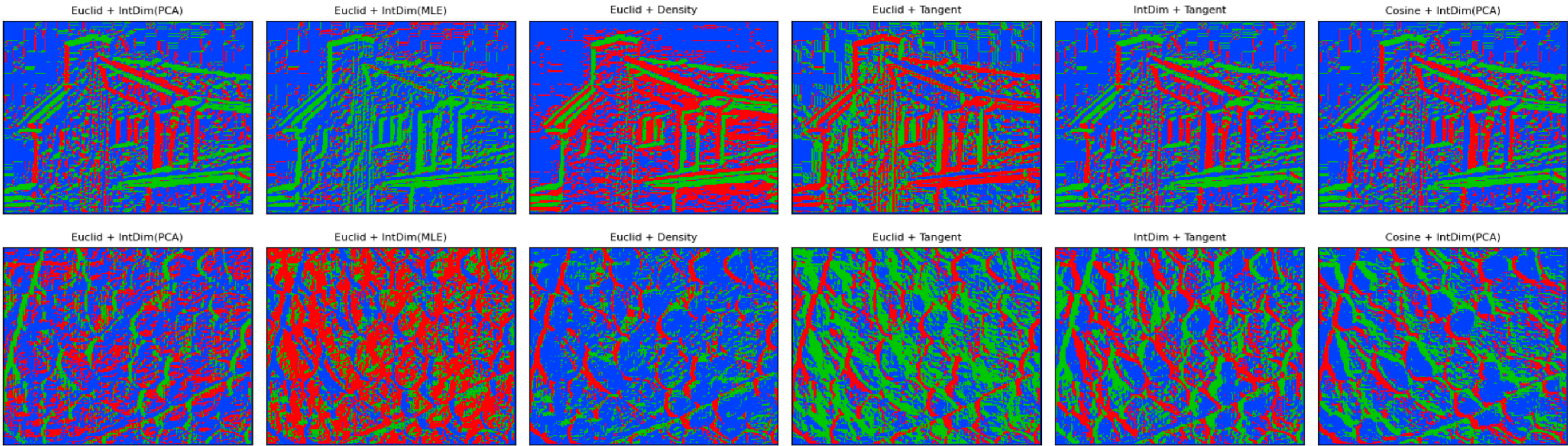


MHC

Intrinsic dimension



Six bifiltration pairs, same PCA anchors, $a = 1.0$



Multiparameter hierarchical clustering

Feature groupings

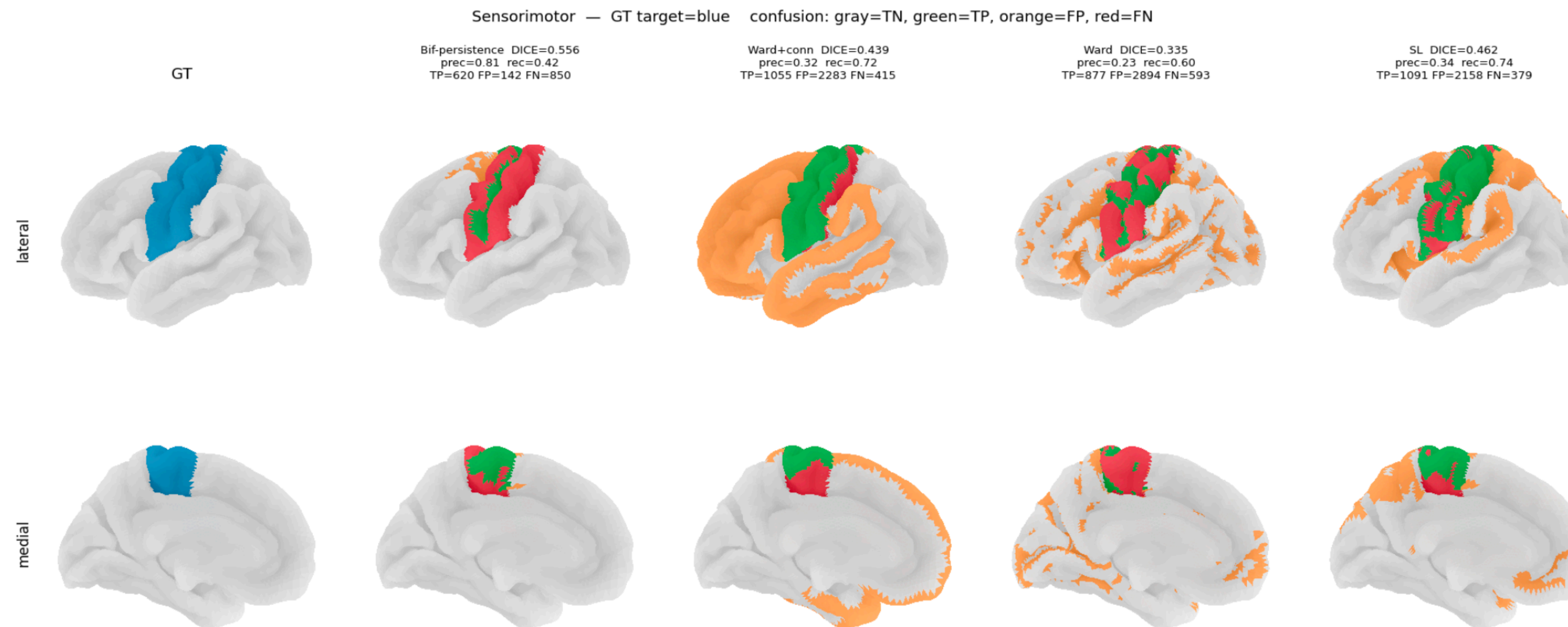


Рис. 5.12: Сенсомоторная полоса. Синий цвет — это разметка. Раскраска: TN — серый, TP — зелёный, FP — оранжевый, FN — красный.

Multiparameter hierarchical clustering

Feature groupings

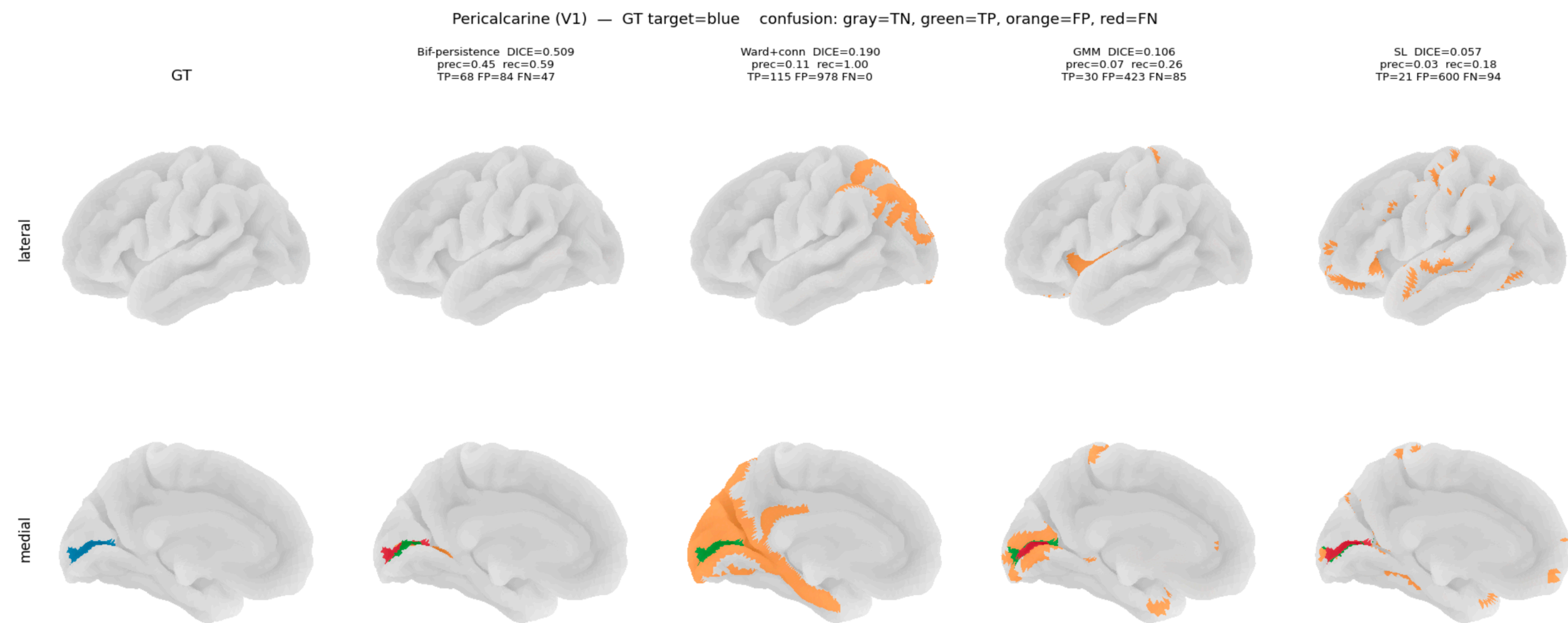


Рис. 5.13: Первичная зрительная кора V1 (pericalcarine). Обозначения как на рис. 5.12.